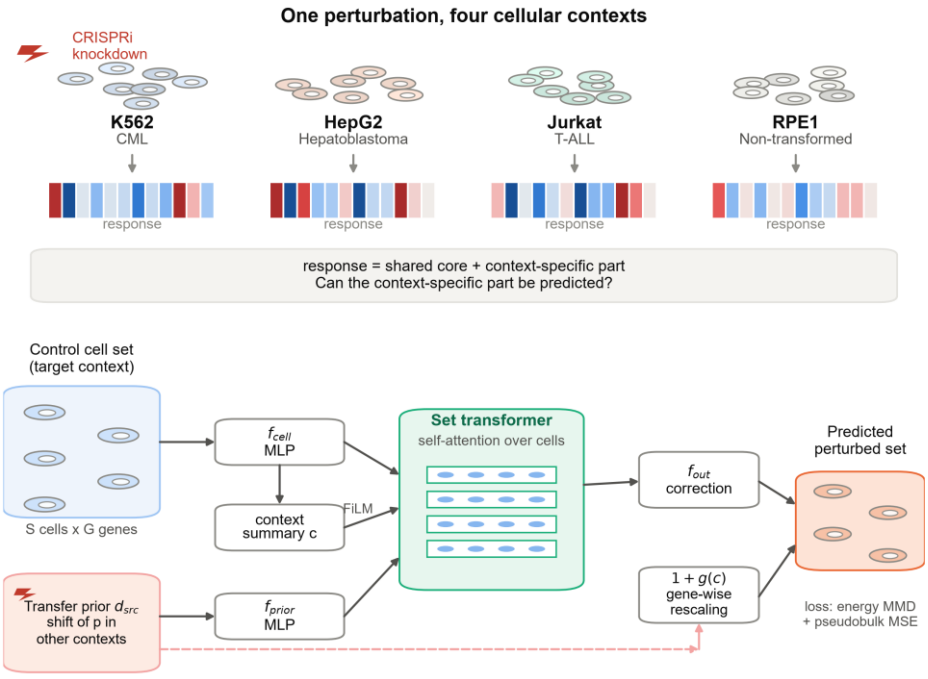
Article

Cancer cell context reshapes genetic perturbation responses: set-based transfer of Perturb-seq effects across leukemia, hepatoblastoma, and non-transformed cells

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Graphical abstract



Highlights

- Responses to the same CRISPRi knockdown are only moderately shared across cell contexts
- CASTER adapts transferred perturbation responses to a context's unperturbed cells
- With partial screening of a context, CASTER outperforms mean baselines
- Knockdown response magnitude tracks DepMap dependency in all three cancer lines

In brief

Adapting perturbation responses measured in one cell type to another, CASTER and a pre-registered benchmark across leukemia, hepatoblastoma, and non-transformed cells show how much cancer context reshapes the transcriptional consequences of losing an essential gene, and how those consequences relate to cancer dependency.

SUMMARY

Genetic perturbation screens are run one cellular context at a time, yet cancers differ in lineage and state. Using CRISPRi Perturb-seq of essential genes in three cancer cell lines and a non-transformed line, we find that transcriptional responses to the same knockdown are only moderately shared across contexts (median correlation 0.31 between cancer lines), with similarity increasing for stronger responses. We introduce CASTER, a set-based transformer that transfers a perturbation's response from other contexts and adapts it to the target context's unperturbed cells. When a fraction of the target context is screened, CASTER improves on mean baselines (Pearson delta 0.28 versus 0.20 for the perturbation mean) and matches a context-adaptive linear model, while in fully unseen contexts the perturbation mean remains hard to beat. Across all three cancer lines, the magnitude of a knockdown's transcriptional response tracked the line's dependency on the gene measured by DepMap (Spearman rho -0.39 to -0.54), and predicted responses retained this dependency information. These results quantify how much cancer context reshapes perturbation responses and provide a reproducible benchmark for context-generalising virtual-cell models.

Keywords: Perturb-seq; CRISPRi; context generalization; virtual cell; set transformer; cancer dependency; DepMap; leukemia; hepatoblastoma; transfer learning

INTRODUCTION

Pooled CRISPR screens with single-cell transcriptomic readouts (Perturb-seq) measure the transcriptome-wide consequence of a genetic perturbation in thousands of cells at once[1–4]. CRISPR interference (CRISPRi)[5,6] has made such screens scalable to every expressed or essential gene[7,8], and they are now used to map gene function, genetic interactions and regulatory programs[8–11]. In cancer research they have identified regulators of immune evasion, drug response and oncogenic variant function[12–16]. Each screen, however, is performed in one cellular context. A tumour is not one context: cancers differ by lineage, driver, ploidy and baseline transcriptional state, and these differences shape how a cell responds to losing a gene[17–20]. Screening every perturbation in every relevant context is not feasible, so computational models that *transfer* perturbation responses across contexts are needed[21,22].

Many models predict single-cell perturbation responses, including autoencoder and latent-arithmetic models[23–27], optimal-transport maps[28], knowledge-graph models[29] and single-cell foundation models[30–32]. Recently, State, a transformer that operates on *sets* of cells and is trained with a distributional loss, was shown to generalise perturbation effects to new contexts better than these approaches on several large datasets[33]. At the same time, careful benchmarks have shown that simple baselines, including the mean response of a perturbation across training contexts and linear models, are hard to beat[34–36]. A community-wide challenge now asks for virtual-cell models that pass a 'Turing test' on held-out contexts[37]. Two questions are therefore open. How much of a perturbation's response is really specific to the context, as opposed to shared? And does that context-specific part carry biological information, for example about which genes a particular cancer depends on?

Here we address both questions in the setting where context matters most for cancer biology: the same essential-gene CRISPRi library profiled in three cancer cell lines from different lineages (K562 chronic myeloid leukaemia, HepG2 hepatoblastoma and Jurkat T cell acute lymphoblastic leukaemia) and in non-transformed hTERT RPE-1 cells[8,38]. We first quantify how similar the responses to the same knockdown are across these contexts. We then introduce CASTER (Context-Adaptive Set Transformer for pErturbation Responses). CASTER represents a perturbation by its observed effect in other contexts (a *transfer prior*) and learns, from the unperturbed cells of the target context alone, how that context rescales and reshapes the response. By construction CASTER starts at the strongest simple baseline and can only move away from it where held-out data support the change. This makes its comparison with baselines a direct test of whether context-specific structure is learnable.

We evaluate CASTER with a pre-registered protocol and a biologically interpretable evaluation suite (PertBench), modelled on the Cell-Eval framework[33]. We test it in under-represented contexts, in completely unseen contexts and in every single-source transfer direction. Finally, we connect perturbation responses to cancer dependency, asking whether the magnitude and context specificity of a knockdown's transcriptional response track CRISPR fitness effects measured independently in the Cancer Dependency Map[39–42]. We report every pre-declared test, including those that do not support our hypotheses.

RESULTS

Responses to the same knockdown are only moderately shared across cellular contexts

We assembled CRISPRi Perturb-seq profiles of essential-gene knockdowns in four human cell lines (Figures 1A and 1B): K562 (chronic myeloid leukemia)[8], HepG2 (hepatoblastoma)[38], Jurkat (T cell acute lymphoblastic leukemia)[38] and the non-transformed, TP53-proficient line hTERT RPE-1[8,43]. We used the knockdown-filtered, uniformly processed release of these data[33] and analysed 196,093 cells: up to 32 cells per perturbation per line and 3,000 controls per line (STAR Methods). After filtering, 1,383 (K562), 1,340 (HepG2), 1,537 (Jurkat) and 1,499 (RPE1) perturbations remained (2,023 distinct genes). Of these, 638 were retained in all four lines and 782 in all three cancer lines. Because on-target efficacy was filtered per line, the set of informative perturbations is itself context dependent (Figure 1B). We modelled 2,000 genes selected on control cells only, excluding every perturbation target, so that neither gene selection nor the trivial on-target signal can inflate any result.

The number of differentially expressed genes (DEGs; Wilcoxon test against held-out controls, FDR < 0.05) varied widely between knockdowns and between contexts. The median was 2 in K562, 5 in HepG2, 2 in Jurkat, 29 in RPE1; 32%, 46%, 32%, 60% of knockdowns, respectively, changed at least 10 genes (Figure S1A). Responses in the non-transformed RPE1 line were thus markedly larger than in the

three cancer lines. This is consistent with an intact p53-dependent stress response to the loss of essential genes in TP53-proficient cells[44,45]. For 706 genes whose knockdown produced >= 10 DEGs in at least two lines, we compared responses across contexts. The shifts were only moderately correlated: median Pearson r was 0.31 between cancer lines and 0.38 between a cancer line and RPE1 (Figures 1D and 1E). DEG sets overlapped little (median Jaccard 0.08 and 0.10; Figure S1C).

The most similar pair was HepG2-RPE1 (median r = 0.52); the other five pairs ranged from 0.28 to 0.33. HepG2 and RPE1 are the two adherent, epithelial-derived lines, whereas K562 and Jurkat grow in suspension, which suggests that cell morphology and growth mode, and not transformation status alone, shape the shared response. Similarity rose with response strength, from 0.31 for knockdowns with >= 10 DEGs to 0.45 for those with >= 200 DEGs in both cancer lines (Figure S1D). Much of the apparent context specificity of weak responses is therefore measurement noise, whereas strong responses share a larger core. A useful context-transfer model must therefore capture the shared core and predict at least part of a substantial context-specific component.

CASTER adapts transferred responses to the target context

These observations motivated CASTER (Figure 1C). Like State[33], CASTER operates on *sets* of cells and is trained with a distributional (energy-distance MMD) loss[46,47], so it models a perturbed population rather than a single average cell. It differs in how it represents the perturbation and the context. A perturbation is represented by its *transfer prior*: the mean shift it produced in the other contexts in which it was observed. The target context is never part of this prior. The context is represented only by a set of its unperturbed cells, which a transformer encoder reads through self-attention. FiLM conditioning[48] lets the same prior act differently in each context. The output adds a context-dependent, gene-wise rescaling of the prior to a cell-resolved correction. Both are initialised at zero, so an untrained CASTER is exactly the PerturbMean baseline, and any departure must be earned on held-out validation data (STAR Methods).

In simulations where the context-specific component was pure noise, CASTER stayed at the PerturbMean solution (Pearson delta 0.67 versus 0.68). When the context-specific component depended on each gene's baseline expression in that context, which is observable from control cells, CASTER recovered it (Pearson delta 0.84 versus 0.49; Figure S4). The architecture can therefore learn context structure when it exists and does not invent it when it does not.

On the K562 under-represented-context fold, validation loss and held-out accuracy changed with the size of the cell set the model reads (Figures 1F and 1G). Pearson delta was 0.178 at S = 1, 0.160 at S = 8, 0.189 at S = 32. Removing self-attention gave 0.181, removing FiLM 0.189, and replacing the transfer prior with a learned perturbation embedding 0.095. Adding an identity embedding to the prior gave 0.185. Paired comparisons of the full model with each variant (Holm-adjusted Wilcoxon) gave: single cells (S = 1): full model better on Pearson delta, DE AUPRC; variant better on discrimination; no attention: full model better on Pearson delta, discrimination, DE AUPRC; no FiLM: no significant advantage; variant better on Pearson delta; no transfer prior: full model better on Pearson delta, discrimination, DE AUPRC; added ID embedding: full model better on Pearson delta, DE AUPRC; S = 8: full model better on Pearson delta, discrimination, DE AUPRC. The transfer prior is by far the most important component. The set-level attention adds a small but significant improvement. A single-cell model (S = 1) discriminates perturbations better, at the cost of the distributional (set-level) objective.

Benchmarking context transfer in under-represented contexts

We first evaluated the common practical setting in which a new context has been screened with only a fraction of a library (Figure 2A)[33]. For each of the four lines in turn, 25% of its eligible perturbations were used for training, 5% for model selection and 70% for testing; the other three lines were fully observed. We compared CASTER with PerturbMean (a perturbation's mean response in the other contexts), ContextMean (the target context's average response), a context-adaptive linear transfer model fitted on the observed fraction, and a no-change floor. The evaluation suite, PertBench, follows Cell-Eval[33]. It scores expression shifts (Pearson delta, perturbation discrimination), differential expression (DEG precision-recall, top-N DEG overlap, fold-change rank correlation, direction) and effect size (rank correlation of the number of DEGs across perturbations) (Figures 2B-2E; STAR Methods).

Averaged over the four held-out contexts, CASTER reached a Pearson delta of 0.28, a discrimination score of 0.43 and a DE AUPRC of 0.19. The corresponding values were 0.20, 0.37 and 0.14 for PerturbMean, 0.28, 0.42 and 0.18 for linear transfer, and 0.21, 0.00 and 0.10 for ContextMean (Figure 2B). Of the nine pre-declared comparisons (H1), 6 favoured CASTER after Holm correction, and 0 significantly favoured a baseline. H1 was therefore supported. DE-based metrics use the amended predicted-side test (AMENDMENT-002, STAR Methods). The pre-registered Wilcoxon test on predicted cells proved invalid: it called a mean of 1333 DEGs per CASTER prediction, versus 61 observed. Under the pre-registered test, 6 of nine comparisons favoured CASTER and 0 favoured a baseline (H1 supported; Table S1); the conclusion from the expression-level metrics does not depend on this choice. Against PerturbMean: Pearson delta: 66% of perturbations improved, Holm $p = 4.0 \times 10^{-152}$; discrimination: 46% of perturbations improved, Holm p = 1.000; DE AUPRC: 67% of perturbations improved, Holm $p = 6.1 \times 10^{-103}$. Against Linear: Pearson delta: 50% of perturbations improved, Holm p = 1.000; discrimination: 50% of perturbations improved, Holm p = 0.117; DE AUPRC: 56% of perturbations improved, Holm $p = 1.6 \times 10^{-17}$. Against ContextMean: Pearson delta: 67% of perturbations improved, Holm $p = 2.0 \times 10^{-149}$; discrimination: 80% of perturbations improved, Holm p < 1e-300; DE AUPRC: 77% of perturbations improved, Holm $p = 1.8 \times 10^{-222}$.

Per context, CASTER and PerturbMean reached Pearson deltas of K562 0.20 versus 0.19; HepG2 0.31 versus 0.24; Jurkat 0.20 versus 0.16; RPE1 0.39 versus 0.20 (Figure 2C). Both transfer methods ranked perturbations by effect size well (Spearman rho between predicted and observed DEG

counts 0.64 for CASTER and 0.63 for PerturbMean; Figure 2E). They recovered DEGs far above the no-change and context-mean baselines (Figure 2D). Among the secondary metrics compared with PerturbMean, 2 favoured CASTER and 1 favoured PerturbMean (Table S1).

Where does CASTER gain over the transfer prior? Binning held-out perturbations by the number of observed DEGs showed that the gain grows with response strength in every context (Spearman rho between gain and log response size 0.22 in K562, 0.49 in HepG2, 0.29 in Jurkat, 0.64 in RPE1; Figure S8). For knockdowns with >= 50 DEGs the median gain in Pearson delta was 0.09 (K562), 0.22 (HepG2), 0.09 (Jurkat), 0.33 (RPE1), whereas for knockdowns with < 10 DEGs it was close to zero (0.00, 0.00, 0.01, 0.03). CASTER's corrections therefore act on real, strong, context-specific responses, not on noise. In the zero-shot task, no such gradient appeared (median gain for strong responses 0.01, -0.05, -0.01, -0.01), in line with the absence of an overall advantage.

Zero-shot transfer and the cancer versus non-transformed distinction

We next asked whether responses could be transferred into a context with no perturbation data at all, using only its unperturbed cells (Figure 3A). In this zero-shot setting CASTER achieved a Pearson delta of 0.17 versus 0.20 for PerturbMean (Figure 3B). The pre-declared comparison (H2) was not supported (Pearson delta: CASTER 0.171 vs 0.197, Holm p = 1.000; discrimination: CASTER 0.214 vs 0.363, Holm p = 1.000; DE AUPRC: CASTER 0.107 vs 0.136, Holm p = 1.000). Per context, CASTER and PerturbMean reached Pearson deltas of K562 0.19 versus 0.19; HepG2 0.17 versus 0.25; Jurkat 0.14 versus 0.15; RPE1 0.19 versus 0.20. Without any perturbation data from the target, the corrections CASTER learns between source contexts did not carry over. Its discrimination fell below that of PerturbMean, indicating that the corrections pulled predictions towards a response shared across perturbations rather than adding perturbation-specific information. By Proposition 4 (STAR Methods), this means that the residual of the transfer prior was not predictable from a context's control cells alone at this data scale. Comparing the gain over PerturbMean with and without a 30% observed fraction (Figure 3E) shows that CASTER's advantage depends on a small amount of perturbation data from the target context. This has a practical design implication: screening a modest subset of a library in a new context, as in our under-represented setting, is what makes context adaptation effective.

To test whether transfer depends on how related two contexts are, we transferred every perturbation from a single source line to each other line (12 directions; Figure 3C). With the PerturbMean transfer, per-perturbation accuracy averaged over cancer-to-cancer directions was 0.14, versus 0.17 for directions involving RPE1 (n = 1,204 perturbations, paired Wilcoxon $p = 9.5 \times 10^{-51}$; Figure 3D). H3 was therefore **not supported**. The difference was significant in the *opposite* direction: transfer involving the non-transformed line was more accurate. The effect was driven by the HepG2-RPE1 pair (Pearson delta 0.23 from HepG2 to RPE1 and 0.26 from RPE1 to HepG2, the two highest entries of the matrix). This matches the higher response similarity of these two adherent, epithelial-derived lines (Figure 1D) and the larger, better-resolved responses of RPE1 (Figure S1A). CASTER showed the same pattern (0.14 versus 0.17, $p = 1.4 \times 10^{-56}$), and in single-source transfer it was indistinguishable from PerturbMean in most directions (Figure 3C; Table S6). Transfer accuracy therefore reflects the biological similarity between two contexts, here lineage and growth mode, more than the binary distinction between transformed and non-transformed cells.

Context-specific differential expression, programs and perturbation structure

At the level of individual perturbations, CASTER and the transfer baselines were compared on DE overlap and fold-change agreement (Figures 4A and 4B). Points above the diagonal are perturbations for which CASTER recovered more of the context's actual DEGs. Figure 4C shows the knockdown with the largest CASTER gain over PerturbMean in a cancer line, illustrating a context-specific correction of the transferred response. Predicted shifts of S-phase and G2/M signature scores[49] had mean absolute errors of 0.0619 (S) and 0.0760 (G2/M) for CASTER, 0.0940 (S) and 0.1378 (G2/M) for PerturbMean (Figure 4D). When perturbations were clustered by the similarity of their responses, predicted similarity maps recovered the observed structure with mean adjusted Rand indices (over 2-20 clusters and four contexts) of 0.26 (CASTER), 0.10 (PerturbMean), 0.23 (Linear), 0.02 (ContextMean) (Figure 4E).

Which perturbations does CASTER adapt most? In each cancer line, the knockdowns with the largest gain over PerturbMean (among those with >= 30 DEGs) were dominated by regulators of ribosome biogenesis and ribosomal proteins: NVL in HepG2 (r over DEGs 0.61 versus -0.16); RBM28 in HepG2 (r over DEGs 0.60 versus 0.00); RPL26 in Jurkat (r over DEGs 0.71 versus 0.32); RRS1 in Jurkat (r over DEGs 0.61 versus 0.25); PDCD11 in K562 (r over DEGs 0.66 versus 0.50); RPL6 in K562 (r over DEGs 0.60 versus 0.36) (Figures 4C and S10). Loss of ribosome biogenesis triggers a nucleolar stress response, whose transcriptional output depends on p53 status, MYC activity and growth rate, all of which differ between these lines. The transferred response of these knockdowns is therefore a poor guide to their response in a new context, while the context's own cells, read through CASTER, carry the information needed to correct it. In HepG2, for example, the PerturbMean prediction for NVL knockdown was uncorrelated or anti-correlated with the observed DEGs, whereas CASTER recovered their direction and relative magnitude.

We identified 15 co-regulated gene programs per context from the observed responses and annotated them with Reactome[50]. Predicted program activities correlated with observed ones at mean r = 0.63 (CASTER), 0.51 (PerturbMean), 0.00 (ContextMean) (Figure 5D). Response similarity between knockdowns recovered high-confidence STRING associations[51] with AUROC 0.81 for observed responses, 0.80 for CASTER, 0.75 for PerturbMean (Figure 5F). Predicted responses therefore preserve much of the functional organisation of the perturbations. In an inverse-design task (identifying which perturbation produced an observed response among all candidates in a context), the correct perturbation was within the top 10 for 56% (CASTER),

26% (PerturbMean), 58% (Linear), 0% (ContextMean) of the 100 strongest perturbations (Figure 5E).

Knockdown responses and cancer dependency

Finally, we asked whether perturbation responses carry information about cancer dependency, measured independently by genome-scale CRISPR fitness screens in DepMap (Chronos gene effects)[39,40,52]. In each cancer line, knockdowns of genes on which the line depends more strongly (more negative Chronos score) were expected to produce larger transcriptional responses (H4). Spearman correlations between log(1 + # DEGs) and Chronos were K562 rho = -0.45 (n = 1,325, Holm p = 7.3×10^{-67}); HepG2 rho = -0.54 (n = 1,275, Holm p = 1.0×10^{-95}); Jurkat rho = -0.39 (n = 1,465, Holm p = 6.8×10^{-54}) (Figure 5A). H4 was supported: response magnitude tracks dependency. Restricting to genes profiled in all three cancer lines, we then asked whether the *context-specific* part of the response (a line's response minus the mean of the other two) tracks the context-specific part of the dependency (H4b): K562 rho = -0.12 (Holm p = 0.004); HepG2 rho = -0.05 (Holm p = 0.318); Jurkat rho = -0.05 (Holm p = 0.318) (Figure 5B). H4b was not supported by the pre-declared rule (significant negative correlation in at least two lines); only K562 reached significance, with a weak effect. In these data, which genes a particular line depends on more than the others is at most weakly reflected in how strongly it responds to their knockdown. Dependency differences between lines are small for most essential genes[53–55], which limits the power of this test.

The dependency-response coupling itself was preserved in predicted responses. Predicted response magnitude correlated with Chronos at rho = -0.41 (K562), -0.52 (HepG2), -0.33 (Jurkat) for CASTER, compared with -0.44, -0.54, -0.43 for observed responses and -0.51, -0.46, -0.33 for PerturbMean. ContextMean, which ignores perturbation identity, lost it (Figure S12). A ridge model trained to predict Chronos from *observed* transcriptional shifts (10-fold cross-validation over held-out perturbations) reached a mean Pearson r of 0.57 across the three cancer lines. Applied unchanged to *predicted* shifts, it reached 0.50 (CASTER), 0.47 (PerturbMean), 0.52 (Linear), -0.02 (ContextMean) (Figure 5C; Figure S5). Transferred responses therefore retain part of the dependency information carried by measured responses. This connects in silico perturbation to cancer target prioritisation[41,42], a use that the reference State model demonstrated for cell viability[33].

Summary of pre-registered outcomes

Table 1 lists every pre-declared hypothesis, its test and its outcome.

Table 1. Pre-registered hypotheses and outcomes

	Hypothesis	Test and decision rule	Outcome
H1	Under-represented context: CASTER > PerturbMean, ContextMean, Linear (3 primary metrics)	One-sided paired Wilcoxon, Holm (9 tests); >= 6 significant, none opposite	Supported (6/9; 0 opposite). Pre-registered DE test: 6/9
H2	Zero-shot context: CASTER > PerturbMean (3 primary metrics)	One-sided Wilcoxon, Holm (3 tests)	Not supported; PerturbMean significantly better on all three
H3	Transfer more accurate among cancer lines than with RPE1	Two-sided paired Wilcoxon	Not supported; opposite direction (p = 9.5×10^{-51})
H4	Response magnitude correlates negatively with Chronos (per cancer line)	Spearman, Holm (3 lines)	Supported in all three lines (rho -0.39 to -0.54)
H4b	Context-specific response tracks context-specific dependency	Spearman, Holm (3 lines)	Not supported (significant in K562 only, rho = -0.12)
H5	Predicted shifts retain dependency information	Exploratory; 10-fold CV ridge	Retained (CASTER r = 0.50 vs observed 0.57)

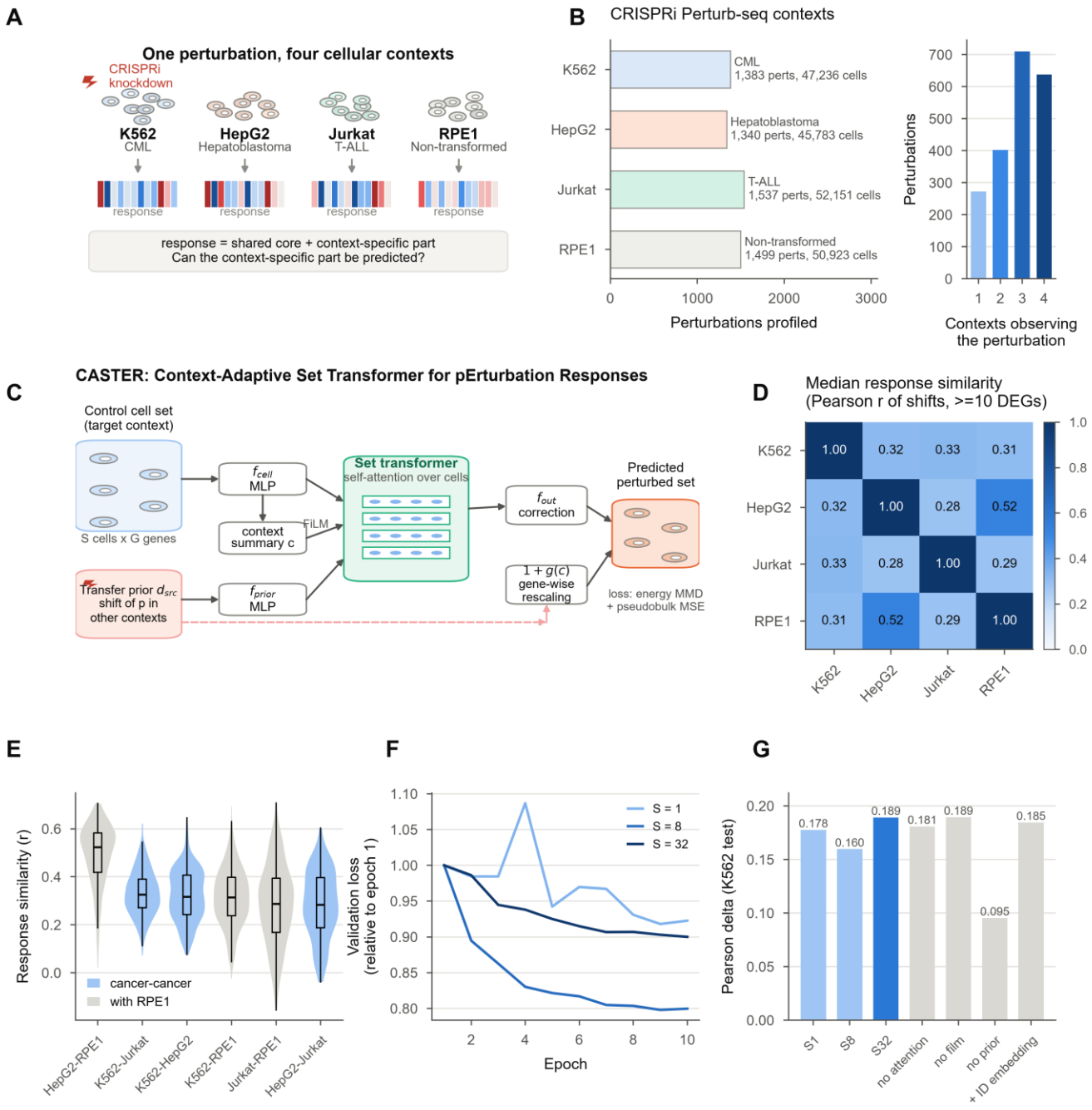


Figure 1. Context reshapes the response to the same genetic perturbation; the CASTER model

(A) The same CRISPRi knockdown, profiled in three cancer contexts and a non-transformed context, produces a shared core response plus a context-specific component.

(B) Perturbations retained after on-target knockdown filtering per context (left), and the number of contexts in which each perturbation is retained (right).

(C) CASTER architecture. A set of unperturbed target-context cells is encoded by an MLP and read by a bidirectional set transformer. The perturbation enters through its transfer prior (mean shift in other contexts), modulated by the context summary (FiLM). The output adds a context-dependent gene-wise rescaling of the prior and a cell-resolved correction; both start at zero.

(D) Median Pearson correlation of observed response shifts between contexts (perturbations with >= 10 DEGs in both).

(E) Per-perturbation response similarity for each pair of contexts; cancer-cancer pairs in blue, pairs involving RPE1 in grey.

(F) Validation loss (relative to epoch 1) for cell-set sizes S = 1-32 on the K562 fold.

(G) Held-out Pearson delta for set sizes and architectural ablations on the K562 fold.

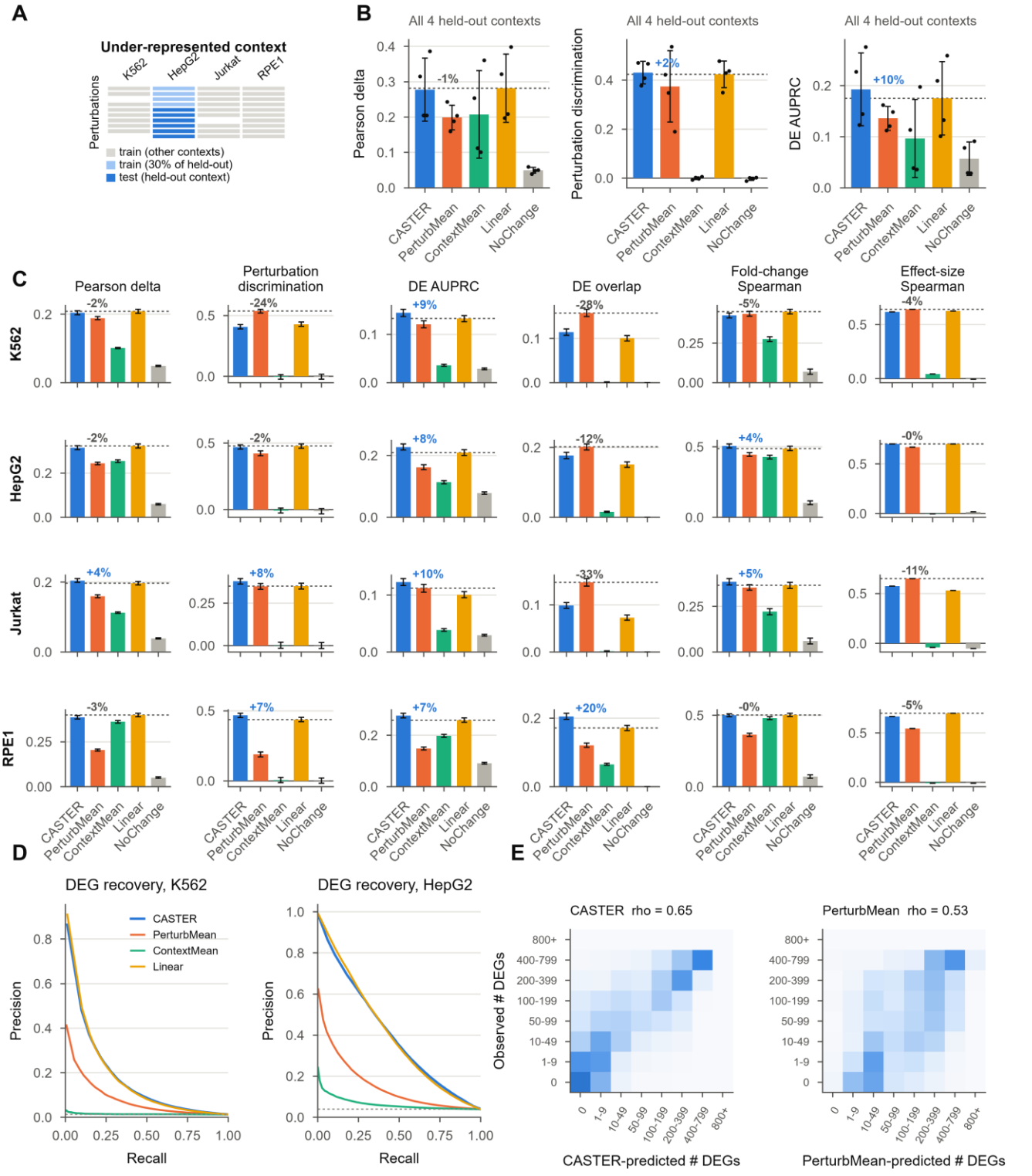


Figure 2. Context transfer in under-represented contexts

(A) Task: for each held-out context, 25% of its perturbations train, 5% validate and 70% test; other contexts are fully observed.

(B) Primary metrics averaged over the four held-out contexts. Bars, mean; error bars, SD over contexts; dots, contexts. Dashed line, best baseline; percentages, relative difference of CASTER from it.

(C) All PertBench metrics per held-out context (error bars, SEM over perturbations; effect-size Spearman is a single value per context).

(D) Precision-recall curves for recovering observed DEGs (pooled over perturbations) in K562 and HepG2, using amended predicted-side p values (AMENDMENT-002); dashed line, base rate.

(E) Observed versus predicted number of DEGs per perturbation (row-normalised), all contexts.

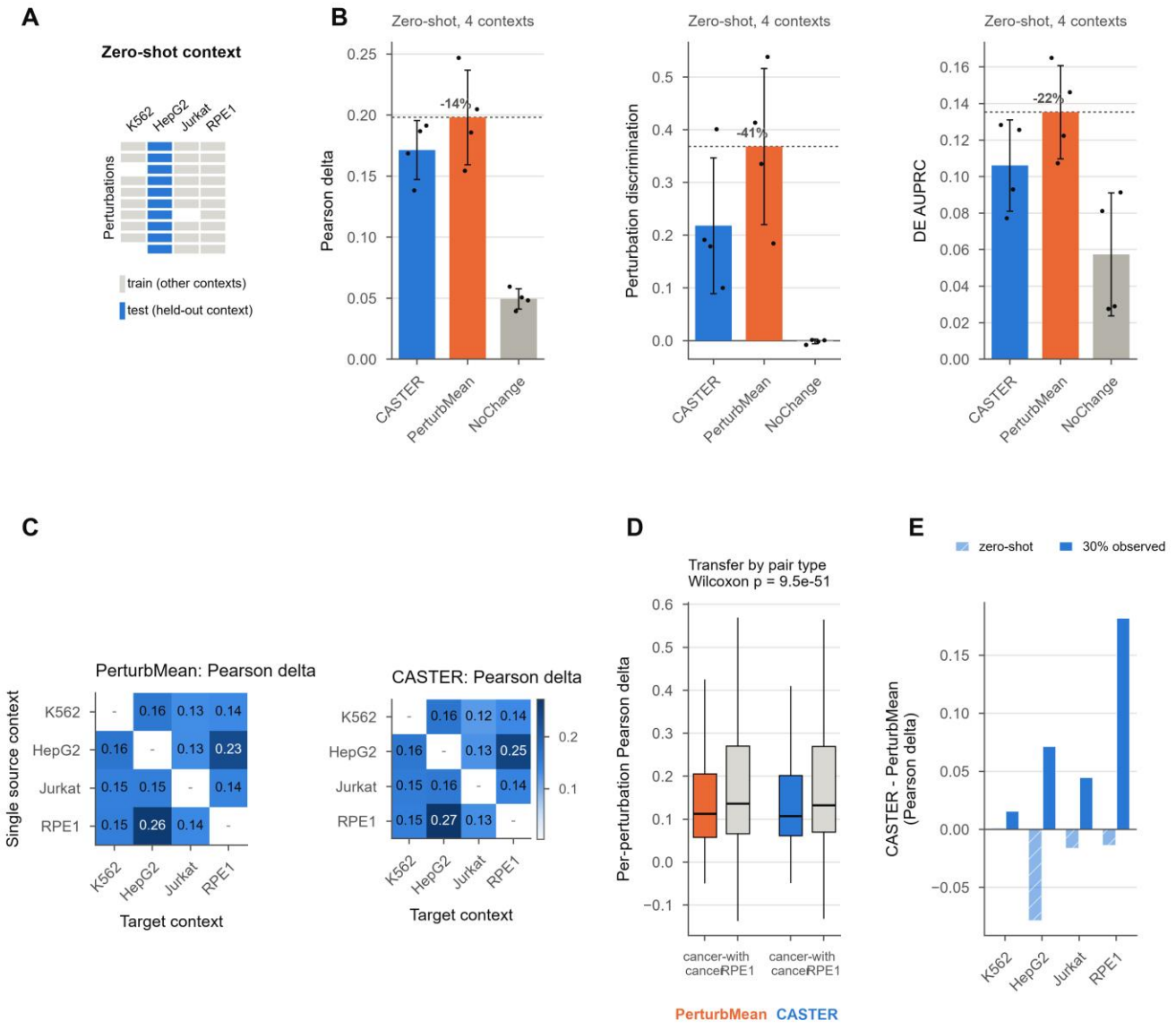


Figure 3. Zero-shot transfer and transfer between cancer and non-transformed contexts

(A) Zero-shot task: no perturbed cells of the held-out context are used.

(B) Primary metrics in the zero-shot task (mean over contexts; dots, contexts).

(C) Single-source transfer: Pearson delta for every source (rows) and target (columns) pair.

(D) Per-perturbation transfer accuracy averaged over cancer-to-cancer directions versus directions involving RPE1. Boxes, interquartile range; whiskers, 1.5x IQR; p value, two-sided paired Wilcoxon (H3).

(E) CASTER minus PerturbMean (Pearson delta) per context, with no (zero-shot) or 30% observed perturbations of the target.

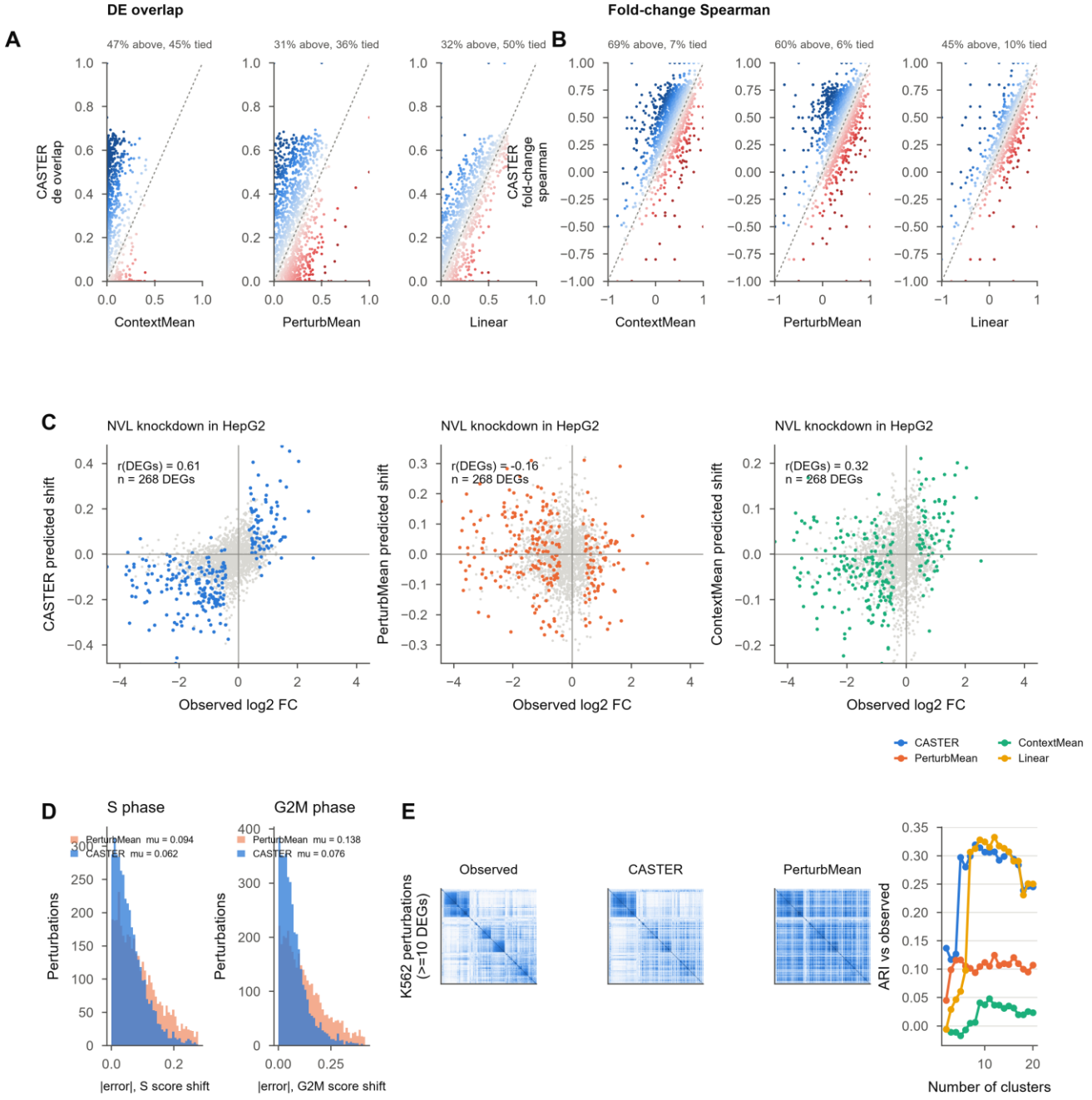


Figure 4. Context-specific differential expression, cell cycle and perturbation structure

(A and B) Per held-out perturbation, CASTER (y) versus each baseline (x) for DE overlap (A) and fold-change Spearman (B); colour, difference (blue, CASTER better).

(C) Observed log2 fold change versus predicted pseudobulk shift for the knockdown with the largest CASTER gain over PerturbMean in a cancer context (coloured, observed DEGs; r over DEGs).

(D) Absolute error of predicted S-phase and G2/M signature shifts.

(E) Perturbation-similarity (cosine) maps for K562 perturbations with >= 10 DEGs, ordered by clustering of the observed map, and adjusted Rand index between predicted and observed clusterings (mean over contexts).

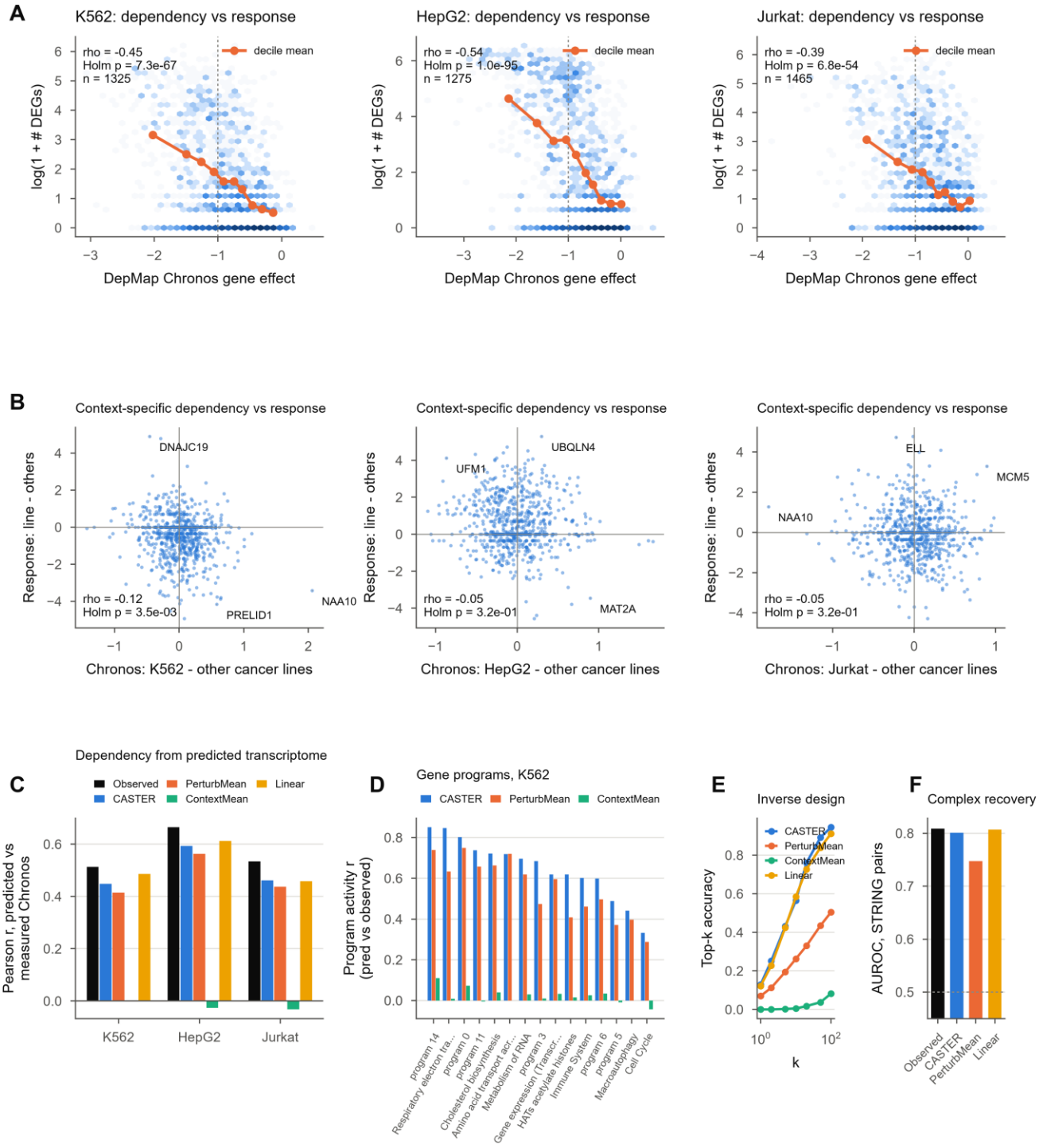


Figure 5. Perturbation responses and cancer dependency

(A) Transcriptional response magnitude (log(1 + # DEGs)) versus DepMap Chronos gene effect of the knocked-down gene in each cancer line (H4). Dashed line, Chronos = -1 (typical essential gene).

(B) Context-specific response versus context-specific dependency (line minus mean of the other two cancer lines; H4b). Labelled, most extreme genes.

(C) Pearson r between measured Chronos and Chronos predicted by a ridge model (trained on observed shifts) from observed or predicted shifts (H5).

(D) Concordance of predicted and observed gene-program activity across perturbations (K562); programs labelled by their top Reactome term when enriched (p < 1e-3).

(E) Inverse design: top-k accuracy of identifying the perturbation that produced an observed response, for the 100 strongest perturbations per context.

(F) AUROC for separating STRING-linked (combined score >= 700) perturbation pairs by response similarity (perturbations with >= 10 DEGs).

DISCUSSION

Perturbation biology is increasingly studied with models that aim to predict the outcome of an experiment before it is run[21,22,37]. For cancer, the context in which a perturbation acts (lineage, driver, transformation state) is central: the same gene can be a vulnerability in one tumour and dispensable in another[39,41,56]. We asked a focused version of this question with a single CRISPRi library profiled in three cancer lines and a non-transformed line. Responses to the same knockdown were only moderately correlated across contexts (median r ~ 0.31 between cancer lines). Similarity increased with response strength, and was highest for the two adherent, epithelial-derived lines. The context-specific component of a perturbation response is therefore large, but part of it reflects measurement noise in weak responses.

CASTER exploits this structure when the target context is partly screened. By anchoring on the transferred response and learning only the context-dependent adaptation, it outperformed the perturbation-mean and context-mean baselines on expression and DE metrics, and recovered gene programs, perturbation-similarity structure and the identity of perturbations far better than the perturbation mean. It matched, but did not exceed, a context-adaptive linear model fitted to the same data. This agrees with recent evidence that well-specified linear models are strong competitors for perturbation prediction[34,35]. The value of the set-based model lies less in accuracy over a tuned linear map than in its distributional, cell-level output and in operating without per-context refitting.

The zero-shot results set a clear boundary. With no perturbation data from the target context, CASTER did not beat the perturbation mean and was worse at discriminating between perturbations. In the language of Proposition 4, the residual of the transfer prior was not predictable from a context's unperturbed cells alone. Unperturbed transcriptomes thus carry little information about how a context will reshape a response, whereas a small number of perturbation measurements in that context carry a lot. For screening design this argues for sparse, library-wide pilot screens in each new context of interest, rather than relying on control profiles alone.

We had expected transfer to be more faithful among cancer lines than to or from the non-transformed line (H3). The opposite held. Transfer involving RPE1 was more accurate, driven by the epithelial pair HepG2-RPE1, and RPE1 knockdowns produced markedly larger responses, consistent with intact p53-dependent stress signalling in TP53-proficient cells[44,45]. Shared lineage and growth mode therefore predicted transferability better than transformation status. For drug discovery, this suggests that the choice of reference context for a perturbation atlas should be guided by lineage and cell state, and that a non-transformed line of the right lineage can be a useful reference for tumour contexts.

The perturbations whose responses CASTER reshaped most were regulators of ribosome biogenesis and ribosomal proteins. This is biologically coherent. Ribosome biogenesis is a cancer dependency that is being explored therapeutically, and impairing it triggers a nucleolar stress response whose outcome depends on p53 status and MYC-driven growth programs[17,18]. These differ between a TP53-mutant leukaemia line, a hepatoblastoma line and TP53-proficient epithelial cells. Context-aware prediction is therefore most valuable precisely for the strong, pathway-level perturbations most relevant to cancer vulnerability, and least needed for weak perturbations, whose apparent context specificity is largely noise (Figure S8).

Compared with State[33], which pretrains on more than 100 million perturbed cells and reports gains over mean baselines across several datasets, CASTER is deliberately small and anchored on an explicit transfer prior. Under that design the comparison with PerturbMean measures the learnable context-specific residual directly (Proposition 4), rather than the combined effect of representation learning and scale. The two approaches are complementary: embeddings learned at scale could replace the control-cell encoder and the transfer prior could be combined with learned perturbation representations. Our results suggest that such models should be evaluated against strong transfer and linear baselines, stratified by response strength, and with predicted-side DE tests that respect the zero inflation of single-cell data.

Linking responses to dependency gave a mixed picture. Response magnitude tracked DepMap fitness effects, and the context-specific part of the response did not track context-specific dependency. Dependencies among the essential genes in this library are largely shared across lines, which limits the variance a context-specific test can exploit. Libraries designed around differential dependencies, such as synthetic-lethal partners of cancer drivers[56,57], would provide a sharper test.

Methodologically, we stress three practices that made these conclusions possible: selecting genes on control cells only and excluding perturbation targets, so that gene selection cannot leak test information; rendering every method's prediction as cells and scoring all of them with the same single-cell DE pipeline; and fixing hypotheses and decision rules before model fitting. Several published comparisons of perturbation models differ in exactly these choices[33–36].

Limitations of the study

This study has several limitations. First, it covers four contexts, three of them cancer lines from different lineages; with so few contexts, context generalisation cannot be separated from lineage generalisation, and conclusions about cancer in general require more cancer contexts. Second, compute and bandwidth limits led us to use at most 32 cells per perturbation and to train compact models on CPUs; State-scale models trained on far larger corpora[33] may extract more context-specific signal. Third, CASTER represents a perturbation by its observed effect elsewhere, so it cannot predict perturbations never observed in any context. Fourth, the dependency analyses rely on CRISPR knockout fitness, whereas the screens use CRISPRi knockdown; knockdown efficacy differs between genes and lines. Fifth, Wilcoxon-based single-cell DE ignores sample structure[58]; we apply it identically to observed and predicted cells, but absolute DEG counts should be interpreted with care. Finally, this work used publicly available data only and did not test any prediction experimentally.

RESOURCE AVAILABILITY

Lead contact

Requests for further information should be directed to the lead contact [to be added].

Materials availability

This study did not generate new materials or new experimental data.

Data and code availability

- All data analysed are public: Replogle et al. 2022 and Nadig et al. 2025 CRISPRi Perturb-seq (Arc Institute knockdown-filtered concatenation on Hugging Face), DepMap 24Q4, STRING v12 and Reactome; identifiers are listed in the key resources table.
- All code (data ingestion, CASTER, baselines, PertBench, statistics, figures and this manuscript) is in the canpert repository [URL/DOI to be added on deposition]. Every number in the paper is generated from reports/results.json; every figure panel ships the CSV it is drawn from.
- The pre-registered protocol and its freeze point are in RESEARCH_PROTOCOL.md (git tag protocol-v1.0).

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AUTHOR CONTRIBUTIONS

[To be completed by the authors.]

DECLARATION OF INTERESTS

The authors declare no competing interests. [To be confirmed by the authors.]

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

During this work the authors used Claude (Anthropic) to write analysis code, run the computational experiments, produce figures, and draft and format the manuscript. The authors must review and edit all content and take full responsibility for the publication.

SUPPLEMENTAL INFORMATION

Tables S1-S8 and Figures S1-S12 with their legends follow the STAR Methods.

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STAR★METHODS

Key resources table

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
K562 and RPE1 CRISPRi Perturb-seq	Replogle et al. 2022	Figshare plus article 20029387
HepG2 and Jurkat CRISPRi Perturb-seq	Nadig et al. 2025	GEO: GSE264667
Knockdown-filtered four-line concatenation (log-normalised)	Arc Institute (State)	huggingface.co/datasets/arcinstitute/State-Replogle-Filtered
DepMap CRISPR gene effect (Chronos), release 24Q4	DepMap / Broad Institute	depmap.org/portal
STRING v12.0 human protein links	Szklarczyk et al. 2023	string-db.org
Reactome pathways (GMT)	Milacic et al. 2024	reactome.org
Experimental models: cell lines (data only)		
K562 (CML)	Lozzio and Lozzio 1975	RRID:CVCL_0004
HepG2 (hepatoblastoma)	Aden et al. 1979	RRID:CVCL_0027
Jurkat (T-ALL)	Schneider et al. 1977	RRID:CVCL_0065
hTERT RPE-1 (non-transformed)	Bodnar et al. 1998	RRID:CVCL_4388
Software and algorithms		
CASTER, PertBench and all analyses	This paper	canpert repository (code availability)
PyTorch 2.13 (CPU)	Paszke et al. 2019	pytorch.org
scikit-learn	Pedregosa et al. 2011	scikit-learn.org
SciPy / NumPy	Virtanen et al. 2020; Harris et al. 2020	scipy.org; numpy.org
h5py / anndata format	Virshup et al. 2024	anndata.readthedocs.io
Matplotlib	Hunter 2007	matplotlib.org

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

This study generated no new experimental data. All analyses use published CRISPR interference (CRISPRi) Perturb-seq profiles of four human cell lines: K562 (chronic myeloid leukaemia, BCR-ABL1-positive)[59], HepG2 (hepatoblastoma-derived, historically described as hepatocellular carcinoma)[60,61], Jurkat (T cell acute lymphoblastic leukaemia)[62,63] and hTERT RPE-1 (telomerase-immortalised, non-transformed retinal pigment epithelium)[43]. K562 and RPE1 were profiled by Replogle et al.[8] and HepG2 and Jurkat by Nadig et al.[38] with the same essential-gene CRISPRi library and protocol.

METHOD DETAILS

Data acquisition

We used the four-line concatenation released with the State model[33] (replogle_concat.h5ad). It contains library-size-normalised, log1p-transformed expression of 6,546 genes measured in all four lines. Perturbations and cells were already filtered for on-target knockdown, as described by Adduri et al.[33]: perturbation median residual target expression <= 0.30, cell residual <= 0.50, and >= 30 cells. The file is 30 GB. Because compute and bandwidth were limited, we never downloaded it whole. We read the cell metadata, chose a stratified random subsample (seed 0) **before any expression value was read**, and fetched only those rows with HTTP byte-range requests against the contiguous dense matrix. The subsample is up to 32 cells per perturbation and line, plus 3,000 non-targeting control cells per line. Every fetched row was checked to be finite, non-negative and in the log1p range, and the set of fetched rows was checked to equal the selected set exactly. Rows fetched twice by overlapping download passes were byte-identical and were kept once.

Gene space

We modelled 2,000 genes. They were chosen as the top genes by bin-normalised dispersion (20 mean bins, Seurat v1 procedure[64]) computed on the **pooled control cells only**, after excluding every gene that is a perturbation target in any line (1850 genes). Because no perturbed cell enters gene selection, held-out perturbation effects cannot leak through the choice of genes. Because knocked-down targets are excluded, the trivial on-target signal cannot inflate any metric.

Problem formulation

Let $\mathcal{X}_\ell^0$ be the distribution of unperturbed cells in context (cell line) ℓ, and $\mathcal{X}_\ell^p$ the distribution after perturbation p. We write the pseudobulk shift of p in ℓ as

$$\delta_{\ell,p} = \mathbb{E}[\mathcal{X}_\ell^p] - \mathbb{E}[\mathcal{X}_\ell^0] = \bar{\delta}_p + \varepsilon_{\ell,p} \quad (1)$$

Here $\bar{\delta}_p$ is the shared component (the mean over contexts) and $\varepsilon_{\ell,p}$ the context-specific component. Context generalisation asks for $\mathcal{X}_t^p$ in a target context t, given control cells of t, observations of p in other contexts and, optionally, a small set of other perturbations observed in t. PerturbMean estimates only $\bar{\delta}_p$. Any method that beats it must recover part of $\varepsilon_{t,p}$.

CASTER architecture

CASTER (Context-Adaptive Set Transformer for pErturbation Responses) maps a set of S control cells $X^0 \in \mathbb{R}^{S \times G}$ from the target context, together with a perturbation, to a set of S predicted perturbed cells. It has five parts.

Transfer prior. The perturbation is represented by its observed shift in the *other* contexts. For an item in context ℓ with source contexts $\mathcal{C}$:

$$d_{\ell,p} = \frac{1}{|\mathcal{C}_p \setminus \{\ell\}|} \sum_{c \in \mathcal{C}_p \setminus \{\ell\}} \hat{\delta}_{c,p} \quad (2)$$

$\mathcal{C}_p$ is the set of source contexts that observed p, and $\hat{\delta}_{c,p}$ is the empirical shift relative to the control mean of c. The target context never contributes to any prior. When no other source context observed p (always the case in single-source transfer), the prior is computed from a random half of p's cells, and the model is trained against the disjoint other half.

Encoders. A 4-layer GELU[65] MLP f_{cell} embeds each control cell into $\mathbb{R}^h$ ($h = 256$). A 2-layer MLP f_{prior} embeds the prior. A context summary c is the mean of the cell embeddings, passed through a 2-layer MLP that outputs FiLM[48] parameters (γ, β):

$$H = f_{cell}(X^0) + h_p + \gamma(c) \odot h_p + \beta(c), \quad h_p = f_{prior}(d_{t,p}) \quad (3)$$

Set transformer. H passes through a 4-layer, 8-head, pre-norm, bidirectional transformer encoder[66,67] without positional encodings. The model is therefore permutation-equivariant over cells[68], which our tests verify.

Output. The prediction adds a context-dependent, gene-wise rescaling of the transferred shift and a cell-resolved correction:

$$\hat{X} = X^0 + \mathbf{1}\,[(1 + g(c)) \odot d_{t,p}]^\top + f_{out}(f_{cell}(X^0) + f_{ST}(H)) \quad (4)$$

Here g is linear in c, and f_{out} is a LayerNorm[69] followed by a linear map. Both g and the output layer of f_{out} are initialised at zero. **Training therefore starts exactly at the PerturbMean solution** and leaves it only where the validation objective improves. The ablations remove the transformer (per-cell MLP), FiLM, or the prior (a learned one-hot perturbation embedding instead), or add a one-hot ID embedding to the prior.

Training objective

For a predicted set $\hat{X}$ and an observed set X of the same perturbation and context, the loss is the squared maximum mean discrepancy[46] with the energy-distance kernel[47,70], plus a pseudobulk term:

$$\mathcal{L} = \frac{2}{S^2}\sum_{i,j}\|\hat{x}_i - x_j\| - \frac{1}{S^2}\sum_{i,j}\|\hat{x}_i - \hat{x}_j\| - \frac{1}{S^2}\sum_{i,j}\|x_i - x_j\| + \lambda\,\|\bar{\hat{x}} - \bar{x}\|_2^2 \quad (5)$$

We used $\lambda = 1$ and $S = 32$. Training used AdamW[71] (learning rate 3e-4, weight decay 1e-4, batch 32 sets), gradient clipping at 1.0, a 5% linear warm-up and cosine decay. The number of epochs was 20 for the context tasks and 10 for ablations and single-source transfer. The checkpoint with the lowest validation loss was kept. In the under-represented task, validation uses the 5% validation perturbations of the target. In the zero-shot and transfer tasks, it uses 5% of source-context items, so no target perturbed cell is ever used. All runs used CPU only (PyTorch[72]).

Baselines

Every method returns a set of predicted cells, so that single-cell DE metrics are computed identically for all methods. A pseudobulk prediction $\hat{\delta}$ is rendered as cells by adding it to the same control cells given to CASTER.

- **NoChange**: $\hat{\delta} = 0$; a floor.
- **PerturbMean**: $\hat{\delta}_{t,p} = d_{t,p}$, the mean shift of p over the source contexts that observed it.
- **ContextMean**: the mean shift of the target's training perturbations, the same for every perturbation[35].
- **Linear transfer**: ridge regression[73] (penalty chosen by efficient leave-one-out cross-validation over 13 values) from a 64-component PCA of the concatenated source shifts of p to its target shift. It is fitted on the target's support perturbations; a missing source is imputed with the mean over the observing sources. This is a strong, explicitly context-adaptive linear model of the kind recently shown to match deep models[34,35].

Tasks and splits

Under-represented context (T1). For each target line t (four folds), the eligible perturbations are those observed in t and in at least one other line. They were shuffled (seed 0) and split 25% training, 5% validation and 70% test. All perturbations of the three other lines are training data. **Zero-shot context (T2).** No perturbed cell of t is used. **Single-source transfer (T3).** For each ordered pair (source s, target t), training uses s only and the test set is every perturbation observed in both. Unperturbed cells of every line, including t, are available to all methods, as in Adduri et al.[33]. Control cells of each line were split 50/50 into a model-input pool and a disjoint DE-reference pool.

Gene programs, similarity and cell-cycle analyses

Gene programs were defined per line by K-means[74] (15 programs, 10 restarts) on the z-scored observed shift matrix (perturbations x genes), clustering genes by their response profiles. Programs were annotated with their most enriched Reactome pathway[50] (hypergeometric test, background = modelled genes). Program activity is the mean shift over program genes. Perturbation-similarity structure used average-linkage hierarchical clustering of cosine distances between shifts of perturbations with >= 10 observed DEGs, compared with the adjusted Rand index[75]. Recovery of functional relationships used STRING v12 pairs with combined score >= 700[51], scored by effect cosine similarity (AUROC). Cell-cycle shifts are the mean shift over the Tirosh et al. S-phase and G2/M signature genes[49] present in the gene space.

Dependency analyses

Chronos gene effects[40] (DepMap 24Q4[39,52]) were taken for K562 (ACH-000551), HepG2 (ACH-000739) and Jurkat (ACH-000995); RPE1 is not profiled. Response magnitude is $\log(1 + n_{DEG})$. The context-specific response of a knockdown in line ℓ is its magnitude minus the mean over the other two cancer lines, and the context-specific dependency is defined the same way on Chronos scores. For dependency prediction (H5), a ridge model was trained, within 10-fold cross-validation over the held-out perturbations of each cancer line, to map *observed* shifts to Chronos. It was then applied unchanged to the shifts predicted by each method for the held-out fold.

Formal definition of baseline models

Let $\bar{x}^0_c$ be the mean of the model-input control cells of context c, and $\bar{x}^p_c$ the mean of the observed cells of perturbation p in c. The observed shift is $\hat{\delta}_{c,p} = \bar{x}^p_c - \bar{x}^0_c$. For the target context t, let S_t be its support perturbations (training + validation) and $\mathcal{C}$ the source contexts.

PerturbMean averages the shift of p over the source contexts that observed it:

$$\hat{\delta}^{\mathrm{PM}}_{t,p} = \frac{1}{|\mathcal{C}_p|} \sum_{c \in \mathcal{C}_p} \hat{\delta}_{c,p} \tag{6}$$

ContextMean predicts the same shift for every perturbation of t, namely the mean shift of its support perturbations:

$$\hat{\delta}^{\mathrm{CM}}_{t,p} = \frac{1}{|S_t|} \sum_{q \in S_t} \hat{\delta}_{t,q} \tag{7}$$

Linear transfer. Each perturbation is described by the concatenation of its source shifts, $u_p = [\hat{\delta}_{c_1,p}; \ldots; \hat{\delta}_{c_K,p}] \in \mathbb{R}^{KG}$. A source that did not observe p contributes $\hat{\delta}^{\mathrm{PM}}_{t,p}$. The features are reduced by PCA, fitted on source data only, to $k = \min(64, |S_t| - 1)$ components $v_p = W^\top (u_p - \bar{u})$. A ridge map $B \in \mathbb{R}^{k \times G}$ is then fitted on the support perturbations:

$$\hat{B} = \arg\min_{B,b} \sum_{q \in S_t} \|\hat{\delta}_{t,q} - B^\top v_q - b\|_2^2 + \lambda \|B\|_F^2 = (V^\top V + \lambda I)^{-1} V^\top \tilde{\Delta} \tag{8}$$

Here V stacks the centred v_q and $\tilde{\Delta}$ the centred target shifts. λ is chosen from 13 log-spaced values in $[10^{-2}, 10^4]$ by efficient leave-one-out cross-validation. The prediction is $\hat{\delta}^{\mathrm{Lin}}_{t,p} = \hat{B}^\top v_p + \hat{b}$. **NoChange** predicts $\hat{\delta} = 0$. Every pseudobulk prediction is rendered as cells by adding it to the same control cells, $\hat{X}_{t,p} = X^0_{t,p} + \mathbf{1}\hat{\delta}^\top_{t,p}$.

Formal definition of downstream analyses

Inverse design. For the 100 perturbations with the most observed DEGs in a context, the rank of the correct perturbation among all predictions is

$$\mathrm{rank}_t = \sum_{t'} \mathbb{1}\left[\|\bar{X}_t - \bar{\tilde{X}}_{t'}\|_1 < \|\bar{X}_t - \bar{\tilde{X}}_t\|_1\right], \quad \mathrm{acc@}k = \frac{1}{100} \sum_t \mathbb{1}[\mathrm{rank}_t < k] \tag{9}$$

STRING-pair recovery. For perturbations with >= 10 observed DEGs, each unordered pair (p, q) is labelled positive if STRING v12 reports a combined score >= 700 between their products. It is scored by the cosine similarity of the predicted shifts, $s_{pq} = \cos(\hat{\delta}_p, \hat{\delta}_q)$, and summarised by the AUROC.

Gene programs. Let Z be the observed shift matrix (perturbations x genes), z-scored per gene. K-means with $K = 15$ (10 restarts, seed 0) on the rows of $Z^\top$ assigns each gene to a program k. The activity of program k under perturbation p is $a_{pk} = \frac{1}{|G_k|} \sum_{g \in G_k} \hat{\delta}_{p,g}$. Concordance is the Pearson correlation across perturbations of predicted and observed $a_{\cdot k}$. Programs are annotated by the Reactome pathway with the smallest hypergeometric p value, with the modelled genes as the background and at least 3 overlapping genes.

Perturbation-similarity structure. Cosine-distance matrices of predicted and observed shifts (perturbations with >= 10 DEGs) were clustered by average linkage into $k = 2, \ldots, 20$ clusters, and the clusterings were compared by the adjusted Rand index.

Dependency prediction (H5). Within each cancer line, held-out perturbations with a measured Chronos score were split into 10 folds. A ridge model f (penalty chosen by leave-one-out cross-validation over 13 values in $[10^{-1}, 10^5]$) was fitted on the *observed* shifts of 9 folds and applied to the observed and to each method's predicted shifts of the remaining fold. The score is the Pearson correlation between the out-of-fold predictions $f(\hat{\delta}_p)$ and the measured Chronos scores. Because the regressor never sees a predicted shift during training, any loss of accuracy reflects information missing from the prediction.

Context-specific dependency (H4b). For gene p observed in all three cancer lines, with response magnitude $r_{\ell,p} = \log(1 + n^{\mathrm{DEG}}_{\ell,p})$ and Chronos score $c_{\ell,p}$, the context-specific components are

$$\Delta r_{\ell,p} = r_{\ell,p} - \tfrac{1}{2} \sum_{\ell' \neq \ell} r_{\ell',p}, \qquad \Delta c_{\ell,p} = c_{\ell,p} - \tfrac{1}{2} \sum_{\ell' \neq \ell} c_{\ell',p} \tag{10}$$

H4b tests the Spearman correlation of $\Delta r_{\ell,\cdot}$ and $\Delta c_{\ell,\cdot}$ per line.

Response similarity across contexts. For each pair of contexts and each perturbation with >= 10 DEGs in both, similarity is the Pearson correlation of the two observed shift vectors over the modelled genes. DEG overlap is the Jaccard index of the two DEG sets. The stratified analysis repeats this for minimum response sizes of 10, 25, 50, 100 and 200 DEGs.

Formal properties of CASTER and of context transfer

We state four properties that frame the benchmark results. Notation follows the problem formulation above. P^0_t denotes the control distribution of context t, $d = d_{t,p}$ the transfer prior, and θ the CASTER parameters.

Proposition 1 (CASTER contains PerturbMean)

At initialisation ($g \equiv 0$, and the final linear layer of f_{out} equal to zero), for any control set X^0 and perturbation p, the CASTER prediction is

$$\hat{X} = X^0 + \mathbf{1}\, d^\top_{t,p} \tag{11}$$

Its pseudobulk shift therefore equals the PerturbMean prediction exactly. *Proof.* The output is $X^0 + \mathbf{1}[(1 + g(c)) \odot d]^\top + f_{out}(\cdot)$. With $g = 0$ the middle term is $\mathbf{1}d^\top$, and $f_{out}(z) = Wz + b$ with $W = 0$ and $b = 0$ is identically zero. Because the checkpoint is chosen by validation loss, which includes the epoch-0 model, the selected CASTER can deviate from PerturbMean only when the deviation lowers the validation objective. ■

Proposition 2 (permutation equivariance)

For any permutation matrix $\Pi \in \{0,1\}^{S \times S}$, $\mathrm{CASTER}(\Pi X^0, p) = \Pi\, \mathrm{CASTER}(X^0, p)$. *Proof.* f_{cell} acts row-wise and commutes with Π. The context summary c is a mean over rows and is invariant. The FiLM and prior terms are broadcast identically to every row. Self-attention without positional encodings is permutation-equivariant[67,68]. f_{out} and the residual act row-wise. A composition of equivariant and row-broadcast invariant maps is equivariant. ■ The property is verified numerically in the test suite. It means that predictions depend on the *distribution* of control cells, not on their order.

Proposition 3 (the training loss identifies distributions)

For distributions P, Q on $\mathbb{R}^G$ with finite first moments, the energy distance[47]

$$\mathcal{E}(P,Q) = 2\,\mathbb{E}\|Y-Z\| - \mathbb{E}\|Y-Y'\| - \mathbb{E}\|Z-Z'\|, \quad Y,Y' \sim P,\ Z,Z' \sim Q \qquad (12)$$

is non-negative and equals zero if and only if $P = Q$. It is the squared MMD with the (conditionally negative definite) energy kernel[46,70]. The empirical V-statistic in the loss converges almost surely to $\mathcal{E}$ as $S \to \infty$. Minimising the population loss therefore matches the full predicted and observed perturbed distributions, not only their means. The added pseudobulk term is minimised by the conditional mean, so the combined objective is minimised by any predictor whose distribution equals the observed one, and both terms vanish there. ■

Proposition 4 (when can context transfer beat PerturbMean?)

Decompose the target shift as $\delta_{t,p} = d_{t,p} + r_{t,p}$, where $r_{t,p}$ is the residual of the transfer prior. Let $\mathcal{F}_t$ be the information available to a predictor at test time: the prior, the control cells of t and, in T1, the support perturbations of t. Under squared error, the optimal prediction of $\delta_{t,p}$ is

$$\delta^{\star}_{t,p} = d_{t,p} + \mathbb{E}[r_{t,p} \mid \mathcal{F}_t] \qquad (13)$$

Its risk improves on that of PerturbMean by exactly $\mathbb{E}\|\mathbb{E}[r_{t,p} \mid \mathcal{F}_t]\|^2$. *Proof.* By the orthogonality of conditional expectation,
$\mathbb{E}\|r - h(\mathcal{F})\|^2 = \mathbb{E}\|r - \mathbb{E}[r \mid \mathcal{F}]\|^2 + \mathbb{E}\|\mathbb{E}[r \mid \mathcal{F}] - h(\mathcal{F})\|^2$ for any measurable h. PerturbMean is $h \equiv 0$. ■ Consequently, *no* method, however expressive, can outperform PerturbMean in expectation unless the residual of the transfer prior is predictable from the target's control cells and support perturbations. The comparison between CASTER (and the linear transfer model) and PerturbMean is therefore an empirical estimate of $\|\mathbb{E}[r_{t,p} \mid \mathcal{F}_t]\|$. A null result means that this predictable part is small relative to estimation noise at the available data scale, not that context does not matter. Indeed $\mathbb{E}\|r_{t,p}\|^2$ (the context-specific variance quantified in Figure 1) can be large while its predictable part is small. In the positive-control simulation the residual is constructed to be a function of the control profile, so this bound is attained.

Information available to each method (leakage audit)

Method	Target controls	Target support perts	Source contexts	Target test perts
NoChange	input pool	-	-	never
ContextMean	input pool	yes (mean)	-	never
PerturbMean	input pool	-	yes	never
Linear transfer	input pool	yes (fit)	yes	never
CASTER (T1)	input pool	yes (train/val)	yes	never
CASTER (T2/T3)	input pool	-	yes (val: 5% source)	never
Evaluation	reference pool	-	-	observed cells

Control cells of each context are split once (seed 0) into disjoint input and reference pools. Gene selection uses control cells only. Every perturbation in the held-out test set is excluded from all training and model-selection steps for that context. This is enforced in code and covered by unit tests (split disjointness, control-pool disjointness, deterministic inputs).

Software, computation and reproducibility

All analyses were run on a single workstation (4 CPU cores, 8 GB RAM, no GPU) with Python 3.12, PyTorch 2.13 (CPU build)[72], NumPy[76], SciPy[77], scikit-learn[78], pandas, h5py and Matplotlib[79]. Under-represented and zero-shot folds took about 35-50 min each to train and evaluate; single-source transfer models about 10 min each. Two jobs ran concurrently with two threads each. All randomness (cell subsampling, control-pool splits, perturbation splits, model initialisation, mini-batch sampling and the control cells given to each method) is seeded. Control input sets are derived deterministically from a hash of the perturbation, seed and context, so that every method receives identical inputs. The pipeline is a sequence of command-line steps (data build, observed DE, T1, T2, ablations, transfer, analysis, figures and manuscript), and each step writes versioned artefacts (per-perturbation metric tables, predicted pseudobulk shifts, per-gene DE arrays and training histories). Every number in this paper is read from a single results file generated from those artefacts, and every figure panel writes the table it plots.

The code base has unit and integration tests. They cover Benjamini-Hochberg correction against SciPy, the DE test on simulated shifts, perturbation discrimination at its bounds, the energy-MMD loss (zero for identical sets, positive for shifted ones), permutation equivariance of CASTER, split disjointness and determinism, control-pool disjointness, and an end-to-end synthetic run in which CASTER and PerturbMean must recover planted effects. Data integrity was verified at acquisition: every selected row was fetched exactly once in the final dataset, and 112,310 rows fetched twice by overlapping download passes were confirmed byte-identical before deduplication.

Deviations from the pre-registered protocol

- **AMENDMENT-001** (before any result was seen): to fit the compute budget, training epochs were set to 20 for context tasks and 10 for ablations and transfer, and the set-size ablation was reduced to S in {1, 8, 32}.
- **AMENDMENT-002** (after T1 results were first inspected): the predicted-side DE test was replaced by a moment-matched z-test because the pre-registered Wilcoxon test is invalid for continuous predictions (STAR Methods;

Figure S7). Both versions of every affected test are reported.

- **Data source** (before freezing): the knockdown-filtered four-line release[33] replaced the per-line original files, for uniform processing and because the original repository throttled range requests.
- Seeds 1 and 2 of the under-represented task (declared secondary) were not run for compute reasons.

QUANTIFICATION AND STATISTICAL ANALYSIS

Evaluation suite (PertBench)

For test perturbation t with observed cells X_t, predicted cells $\hat{X}_t$ (same number of cells) and reference controls R (500 cells, disjoint from model inputs):

Pearson delta. The correlation between predicted and observed pseudobulk shifts:

$$\rho_t = \mathrm{corr}(\bar{\hat{X}}_t - \bar{R},\ \bar{X}_t - \bar{R}) \tag{14}$$

Perturbation discrimination. Following Wu et al.[36] and Adduri et al.[33], with r_t the number of other test perturbations whose observed pseudobulk is closer in L1 distance to the prediction than the true one:

$$\mathrm{PDisc}_t = 1 - \frac{2r_t}{T-1}, \qquad r_t = \sum_{t' \neq t} \mathbb{1}\left[\|\bar{\hat{X}}_t - \bar{X}_{t'}\|_1 < \|\bar{\hat{X}}_t - \bar{X}_t\|_1\right] \tag{15}$$

A perfect model scores 1 and chance scores 0. **Observed differential expression (ground truth).** Observed cells of each perturbation are tested against R gene by gene with the Wilcoxon rank-sum test[80,81] (asymptotic, tie-corrected), with Benjamini-Hochberg correction[82] at FDR < 0.05. Observed log2 fold changes are log2 ratios of mean expm1-transformed expression with pseudocount 0.01. The limitations of cell-level DE tests are well known[58,83].

Predicted differential expression (AMENDMENT-002). The pre-registered protocol applied the same Wilcoxon test to predicted cells. Inspection of the results showed that this is invalid for continuous predictions. Adding a shift to a control cell turns each of its exact zeros into a small positive value, and against zero-inflated reference cells the rank test then calls most genes significant whatever the shift. In the HepG2 fold, every shifted method produced 1,283-1,494 'DEGs' per perturbation, versus 0.1 for unshifted control cells. We therefore score predictions with a moment-matched z-test on the predicted pseudobulk shift $\hat{\delta}_g$, using the reference-control standard deviation s_g and the sample sizes of the observed experiment:

$$z_g = \frac{\hat{\delta}_g}{s_g\sqrt{1/n + 1/m}}, \qquad p_g = 2\,\Phi(-|z_g|) \tag{16}$$

Here n is the number of perturbed cells and $m = 500$, with BH adjustment. Predicted fold-change ranks and signs use $\hat{\delta}_g$. Observed ground truth is unchanged. The pre-registered Wilcoxon results are reported in full in Table S1 and flagged as invalid. **DE AUPRC** is the average precision[84,85] for recovering observed DEGs from predicted -log10 adjusted p values. **DE overlap** is

$$\mathrm{Overlap}_t = \frac{|G^{(N)}_{t,\mathrm{true}} \cap G^{(N)}_{t,\mathrm{pred}}|}{N}, \quad N = |G^{\mathrm{DE}}_{t,\mathrm{true}}| \tag{17}$$

Here both sets are ranked by |log2 FC| among significant genes. **Fold-change Spearman** is computed over observed DEGs. **Direction agreement** is the fraction of observed DEGs with the correct predicted sign. **Effect-size Spearman** is the rank correlation, across perturbations within a fold, between the predicted and observed numbers of DEGs.

Statistics

All hypotheses, tests and the decision rule were fixed in a protocol (RESEARCH_PROTOCOL.md, git tag protocol-v1.0) before any model was fitted to real data. H1 used one-sided paired Wilcoxon signed-rank tests over held-out perturbations pooled across the four folds (CASTER > baseline), with Holm correction[86] over nine tests (three baselines x three primary metrics). It was declared supported if at least six were significant and none was significant in the opposite direction. H2 used the same test against PerturbMean on T2 (Holm over three). H3 compared per-perturbation single-source transfer accuracy averaged over cancer-to-cancer pairs with the average over pairs involving RPE1 (two-sided paired Wilcoxon). H4 and H4b used Spearman correlations with Holm correction over the three cancer lines. Exploratory analyses are reported without multiplicity claims. Every number in the text is read from the results.json file written by the analysis pipeline.

SUPPLEMENTAL INFORMATION

Table S1. All paired tests, under-represented context task (one-sided Wilcoxon, CASTER > baseline; Holm within each family)

Baseline	Metric	n	CASTER	Baseline	% better	p	p (Holm)
PerturbMean	pearson_delta	3,841	0.278	0.198	66	5.8×10^{-153}	4.0×10^{-152}
PerturbMean	discrimination	3,841	0.430	0.369	46	0.648	1.000
PerturbMean	de_auprc	2,666	0.198	0.137	67	1.2×10^{-103}	6.1×10^{-103}
ContextMean	pearson_delta	3,841	0.278	0.208	67	3.3×10^{-150}	2.0×10^{-149}
ContextMean	discrimination	3,841	0.430	0.000	80	< 1e-300	< 1e-300
ContextMean	de_auprc	2,666	0.198	0.104	77	2.3×10^{-223}	1.8×10^{-222}
Linear	pearson_delta	3,841	0.278	0.282	50	0.960	1.000
Linear	discrimination	3,841	0.430	0.422	50	0.039	0.117
Linear	de_auprc	2,666	0.198	0.181	56	4.0×10^{-18}	1.6×10^{-17}
PerturbMean	mae	3,841	0.077	0.093	2	1.000	1.000
PerturbMean	de_overlap	2,666	0.152	0.154	31	0.562	1.000
PerturbMean	lfc_spearman	2,094	0.459	0.395	60	2.5×10^{-40}	2.0×10^{-39}
PerturbMean	direction	2,666	0.804	0.756	52	1.7×10^{-83}	1.5×10^{-82}
ContextMean	mae	3,841	0.077	0.082	24	1.000	1.000
ContextMean	de_overlap	2,666	0.152	0.024	47	1.6×10^{-171}	1.8×10^{-170}
ContextMean	lfc_spearman	2,094	0.459	0.365	69	3.8×10^{-100}	3.8×10^{-99}
ContextMean	direction	2,666	0.804	0.701	61	1.2×10^{-183}	1.5×10^{-182}
Linear	mae	3,841	0.077	0.077	47	0.961	1.000
Linear	de_overlap	2,666	0.152	0.127	32	5.1×10^{-28}	3.5×10^{-27}
Linear	lfc_spearman	2,094	0.459	0.456	45	0.266	1.000
Linear	direction	2,666	0.804	0.803	30	0.481	1.000
PerturbMean (pre-reg. Wilcoxon)	de_auprc	2,666	0.039	0.035	73	1.6×10^{-152}	7.8×10^{-152}
ContextMean (pre-reg. Wilcoxon)	de_auprc	2,666	0.039	0.036	85	5.8×10^{-246}	4.6×10^{-245}
Linear (pre-reg. Wilcoxon)	de_auprc	2,666	0.039	0.037	85	5.1×10^{-244}	3.6×10^{-243}

Table S2. CASTER hyperparameters and training settings

Component	Setting
Cell encoder f_cell	4-layer MLP, GELU, 2,000 -> 256 -> 256 -> 256 -> 256
Prior encoder f_prior	2-layer MLP, GELU, 2,000 -> 256 -> 256
Context summary / FiLM	mean of cell embeddings -> 2-layer MLP -> (gamma, beta) in R^256 x R^256
Set transformer	4 layers, 8 heads, d_model 256, feed-forward 512, pre-norm, GELU, dropout 0, no positional encoding
Output	LayerNorm + linear 256 -> 2,000 (zero-initialised); g(c): linear 256 -> 2,000 (zero-init)
Cell-set size S	32 (ablation: 1, 8)
Loss	energy-distance MMD^2 + 1.0 x pseudobulk squared error
Optimiser	AdamW, lr 3e-4, weight decay 1e-4, batch 32 sets, grad-norm clip 1.0
Schedule	5% linear warm-up, cosine decay to 5% of peak
Epochs	20 (T1, T2); 10 (ablations, single-source transfer)
Model selection	lowest validation loss (T1: 5% target perturbations; T2/T3: 5% source items)
Hardware	CPU only (4 cores, 8 GB RAM), 2 threads per job, PyTorch 2.13
Parameters	4.6 M (excluding the one-hot embedding, used only in ablations)

Table S3. Dataset after subsampling and filtering

Context	Cells	Controls	Perturbations	Median cells/pert	Median # DEGs	% with >= 10 DEGs
K562	47,236	3,000	1,383	32	2	32
HepG2	45,783	3,000	1,340	32	5	46
Jurkat	52,151	3,000	1,537	32	2	32
RPE1	50,923	3,000	1,499	32	29	60

Table S4. All metrics per held-out context, under-represented task (mean over held-out perturbations)

Context	Method	Pearson d	Discrim.	MAE	DE AUPRC	DE overlap	FC Spearman	Direction	Effect rho
HepG2	CASTER	0.315	0.470	0.070	0.227	0.177	0.506	0.813	0.703

HepG2	ContextMean	0.254	-0.007	0.076	0.114	0.016	0.428	0.730	-0.003
HepG2	Linear	0.322	0.478	0.069	0.210	0.151	0.488	0.814	0.703
HepG2	NoChange	0.060	-0.012	0.078	0.078	0.000	0.103	0.549	0.017
HepG2	PerturbMean	0.243	0.423	0.086	0.162	0.201	0.445	0.785	0.671
Jurkat	CASTER	0.205	0.378	0.085	0.123	0.099	0.386	0.777	0.575
Jurkat	ContextMean	0.113	0.002	0.088	0.039	0.002	0.221	0.627	-0.040
Jurkat	Linear	0.197	0.349	0.085	0.101	0.073	0.367	0.766	0.529
Jurkat	NoChange	0.039	0.001	0.090	0.029	0.000	0.060	0.516	-0.050
Jurkat	PerturbMean	0.160	0.348	0.098	0.112	0.148	0.354	0.757	0.650
K562	CASTER	0.205	0.408	0.072	0.145	0.115	0.426	0.815	0.617
K562	ContextMean	0.101	-0.003	0.075	0.036	0.001	0.276	0.655	0.043
K562	Linear	0.209	0.429	0.072	0.133	0.101	0.450	0.805	0.626
K562	NoChange	0.049	-0.002	0.076	0.029	0.000	0.069	0.553	-0.005
K562	PerturbMean	0.189	0.537	0.089	0.121	0.158	0.436	0.789	0.643
RPE1	CASTER	0.386	0.468	0.082	0.276	0.205	0.501	0.812	0.668
RPE1	ContextMean	0.361	0.006	0.088	0.198	0.065	0.480	0.773	-0.009
RPE1	Linear	0.399	0.438	0.081	0.259	0.171	0.503	0.823	0.701
RPE1	NoChange	0.050	0.002	0.095	0.092	0.000	0.071	0.541	-0.010
RPE1	PerturbMean	0.204	0.190	0.098	0.148	0.121	0.364	0.710	0.544

Table S5. All metrics per held-out context, zero-shot task

Context	Method	Pearson d	Discrim.	MAE	DE AUPRC	DE overlap	FC Spearman	Direction	Effect rho
HepG2	CASTER	0.169	0.191	0.081	0.128	0.109	0.331	0.715	0.672
HepG2	NoChange	0.060	-0.008	0.079	0.081	0.000	0.099	0.549	0.010
HepG2	PerturbMean	0.247	0.414	0.087	0.165	0.204	0.458	0.784	0.666
Jurkat	CASTER	0.138	0.179	0.089	0.077	0.071	0.305	0.717	0.581
Jurkat	NoChange	0.039	0.001	0.089	0.028	0.000	0.054	0.531	-0.010
Jurkat	PerturbMean	0.155	0.335	0.097	0.107	0.144	0.352	0.756	0.635
K562	CASTER	0.187	0.401	0.081	0.093	0.105	0.409	0.769	0.635
K562	NoChange	0.051	-0.001	0.075	0.029	0.000	0.067	0.552	-0.009
K562	PerturbMean	0.186	0.538	0.089	0.122	0.159	0.416	0.784	0.643
RPE1	CASTER	0.191	0.100	0.091	0.126	0.053	0.316	0.683	0.556
RPE1	NoChange	0.048	0.001	0.095	0.091	0.000	0.063	0.541	0.002
RPE1	PerturbMean	0.205	0.185	0.099	0.146	0.118	0.363	0.715	0.533

Table S6. Single-source transfer (mean over shared perturbations)

Source	Target	Method	Pearson d	Discrim.	DE AUPRC	DE overlap
HepG2	Jurkat	CASTER	0.127	0.153	0.054	0.066
HepG2	Jurkat	PerturbMean	0.132	0.286	0.064	0.112
HepG2	K562	CASTER	0.160	0.287	0.077	0.096
HepG2	K562	PerturbMean	0.162	0.475	0.087	0.134
HepG2	RPE1	CASTER	0.246	0.138	0.154	0.089
HepG2	RPE1	PerturbMean	0.235	0.270	0.150	0.159
Jurkat	HepG2	CASTER	0.157	0.117	0.098	0.090
Jurkat	HepG2	PerturbMean	0.150	0.262	0.096	0.166
Jurkat	K562	CASTER	0.152	0.313	0.090	0.104
Jurkat	K562	PerturbMean	0.148	0.521	0.096	0.139
Jurkat	RPE1	CASTER	0.143	0.068	0.107	0.047
Jurkat	RPE1	PerturbMean	0.144	0.190	0.107	0.136
K562	HepG2	CASTER	0.164	0.066	0.107	0.059
K562	HepG2	PerturbMean	0.161	0.206	0.104	0.145
K562	Jurkat	CASTER	0.119	0.096	0.073	0.057
K562	Jurkat	PerturbMean	0.132	0.274	0.085	0.127
K562	RPE1	CASTER	0.135	0.027	0.118	0.033
K562	RPE1	PerturbMean	0.135	0.103	0.118	0.106
RPE1	HepG2	CASTER	0.272	0.411	0.180	0.201
RPE1	HepG2	PerturbMean	0.262	0.558	0.177	0.230
RPE1	Jurkat	CASTER	0.130	0.211	0.084	0.100
RPE1	Jurkat	PerturbMean	0.136	0.363	0.100	0.140
RPE1	K562	CASTER	0.149	0.283	0.087	0.101
RPE1	K562	PerturbMean	0.150	0.430	0.102	0.140

Table S7. Ablations (under-represented task, K562 held out)

Variant	Pearson d	Discrim.	DE AUPRC	DE overlap	FC Spearman
S1	0.178	0.472	0.123	0.160	0.426
S32	0.189	0.323	0.127	0.093	0.423
S32_noattn	0.181	0.276	0.124	0.089	0.404
S32_nofilm	0.189	0.321	0.128	0.093	0.421
S32_noprior	0.095	-0.003	0.035	0.004	0.225
S32_pertid	0.185	0.314	0.124	0.092	0.416
S8	0.160	0.211	0.100	0.070	0.386

Table S8. Gene programs and their top Reactome enrichment

Context	Program	Genes	Top Reactome pathway	p
HepG2	0	199	Cell Cycle	5.0e-23
HepG2	14	109	Cholesterol biosynthesis	7.5e-07
HepG2	6	83	Metabolism	9.0e-06
HepG2	2	127	mRNA Splicing	3.2e-05
HepG2	10	68	DNA Double Strand Break Response	4.5e-05
HepG2	1	85	Metabolism	4.6e-05
HepG2	5	180	Anchoring of the basal body to the plasma membrane	5.9e-05
HepG2	3	78	Generic Transcription Pathway	6.0e-05
HepG2	4	193	CDC42 GTPase cycle	3.1e-04
HepG2	13	86	Hemostasis	7.6e-04
HepG2	8	133	Protein-protein interactions at synapses	1.1e-03
HepG2	7	161	Immune System	1.8e-03
HepG2	12	186	Transcriptional and post-translational regulation of MITF-M expression	2.0e-03
HepG2	9	142	RSV-host interactions	2.3e-03
HepG2	11	170	tRNA modification in the nucleus and cytosol	2.7e-03
Jurkat	2	62	Cell Cycle, Mitotic	1.1e-17
Jurkat	5	106	Cell Cycle	9.4e-12
Jurkat	8	100	Cholesterol biosynthesis	1.2e-08
Jurkat	7	122	Metabolism of RNA	3.0e-08
Jurkat	11	183	Sensory processing of sound by outer hair cells of the cochlea	1.3e-05
Jurkat	3	271	RNA Polymerase II Transcription	4.2e-05
Jurkat	10	113	Muscle contraction	4.4e-05
Jurkat	9	37	Amino acid transport across the plasma membrane	5.7e-05
Jurkat	1	183	RIPK1-mediated regulated necrosis	3.2e-04
Jurkat	6	151	Metabolism	7.3e-04
Jurkat	4	229	Glutamate and glutamine metabolism	2.1e-03
Jurkat	12	246	Stimuli-sensing channels	5.8e-03
Jurkat	14	125	ER to Golgi Anterograde Transport	0.05
Jurkat	0	13	Signal Transduction	0.34
Jurkat	13	59	Cytokine Signaling in Immune system	0.52
K562	12	55	Cell Cycle	1.4e-14
K562	10	128	Immune System	1.6e-07
K562	13	132	Metabolism of RNA	1.6e-05
K562	7	79	Macroautophagy	3.1e-05
K562	1	129	Cholesterol biosynthesis	4.3e-05
K562	9	48	Amino acid transport across the plasma membrane	1.3e-04
K562	2	152	Respiratory electron transport	3.0e-04
K562	8	172	Gene expression (Transcription)	5.6e-04
K562	4	264	HATs acetylate histones	9.7e-04
K562	14	192	Signaling by FGFR3	1.1e-03
K562	11	104	Anchoring of the basal body to the plasma membrane	2.1e-03
K562	5	108	Biological oxidations	2.6e-03
K562	6	256	SUMOylation of ubiquitinylation proteins	6.7e-03
K562	3	152	Collagen formation	0.02
K562	0	29	(none)	1.00
RPE1	8	283	Cell Cycle	7.2e-28
RPE1	7	70	Asparagine N-linked glycosylation	3.3e-08
RPE1	0	38	Cellular response to chemical stress	7.5e-07

RPE1	12	133	Intraflagellar transport	2.7e-05
RPE1	9	47	Amino acid transport across the plasma membrane	1.2e-04
RPE1	3	109	Nervous system development	4.3e-04
RPE1	10	149	mRNA Splicing	7.3e-04
RPE1	2	234	NR1H2 and NR1H3-mediated signaling	8.3e-04
RPE1	13	186	Axon guidance	9.8e-04
RPE1	4	29	Platelet degranulation	1.9e-03
RPE1	5	155	Maternal to zygotic transition (MZT)	2.0e-03
RPE1	6	165	Negative Regulation of CDH1 Gene Transcription	2.7e-03
RPE1	1	285	ATP-dependent chromatin remodelers	9.5e-03
RPE1	11	93	Extracellular matrix organization	0.01
RPE1	14	24	(none)	1.00

Table S9. Fold sizes and training summary

Task	Held-out context	Train perts	Val perts	Test perts	Selected epoch	Minutes
Under-represented	HepG2	320	64	894	18	34.6
Under-represented	Jurkat	366	73	1,027	18	36.3
Under-represented	K562	327	65	917	18	36.0
Under-represented	RPE1	358	72	1,003	14	31.3
Zero-shot	HepG2	0	0	1,278	15	29.7
Zero-shot	Jurkat	0	0	1,466	6	55.9
Zero-shot	K562	0	0	1,309	17	33.7
Zero-shot	RPE1	0	0	1,433	18	19.3

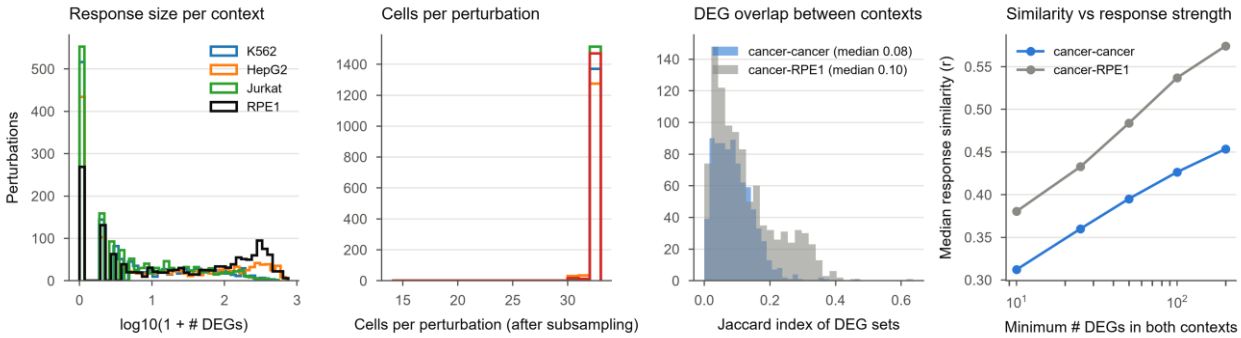


Figure S1. Response sizes and context overlap, related to Figure 1

(A) Distribution of the number of DEGs per perturbation in each context. (B) Cells per perturbation after subsampling. (C) Jaccard overlap of DEG sets between contexts. (D) Median cross-context response similarity as a function of response strength (minimum number of DEGs in both contexts).

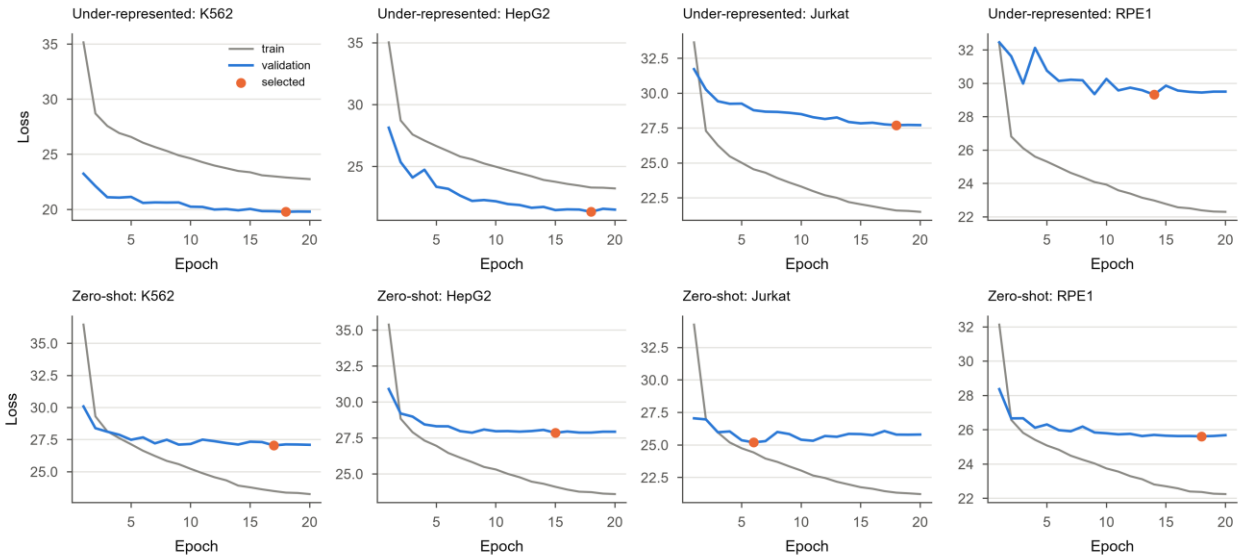


Figure S2. Training curves, related to Figures 2 and 3

Training and validation loss per epoch for every under-represented (top) and zero-shot (bottom) fold; the selected checkpoint is marked.

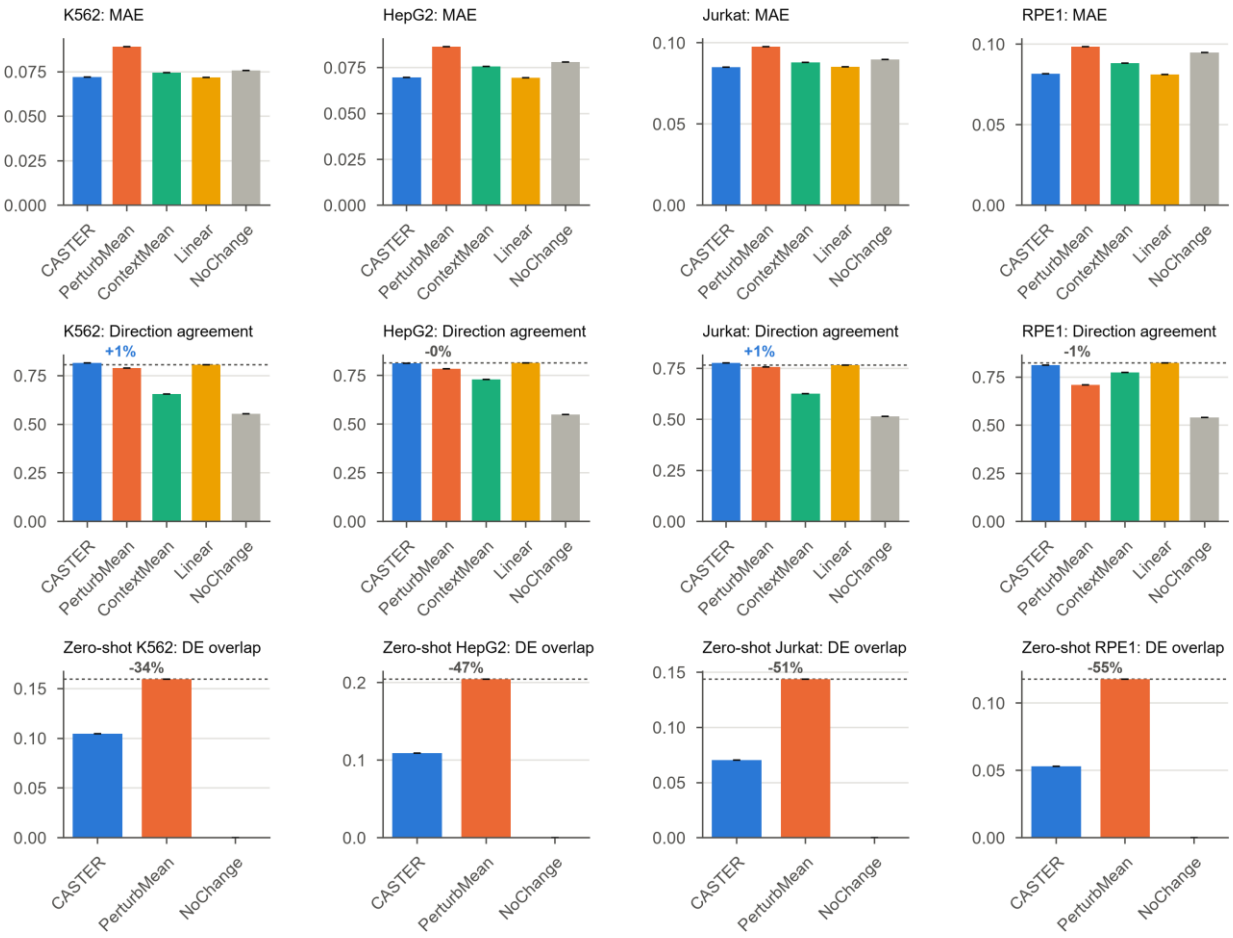


Figure S3. Secondary metrics, related to Figures 2 and 3

Mean absolute error and direction agreement per context (under-represented task), and DE overlap per context in the zero-shot task.

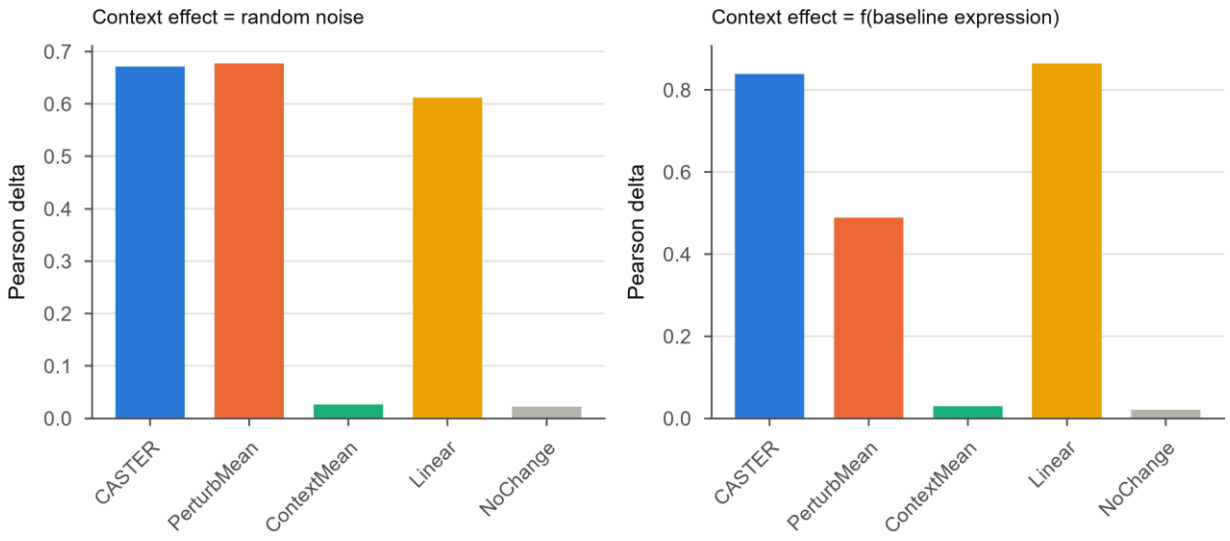


Figure S4. Positive-control simulation, related to Figure 1

Synthetic four-context data in which the context-specific component is random noise (left) or a function of baseline expression observable from control cells (right).

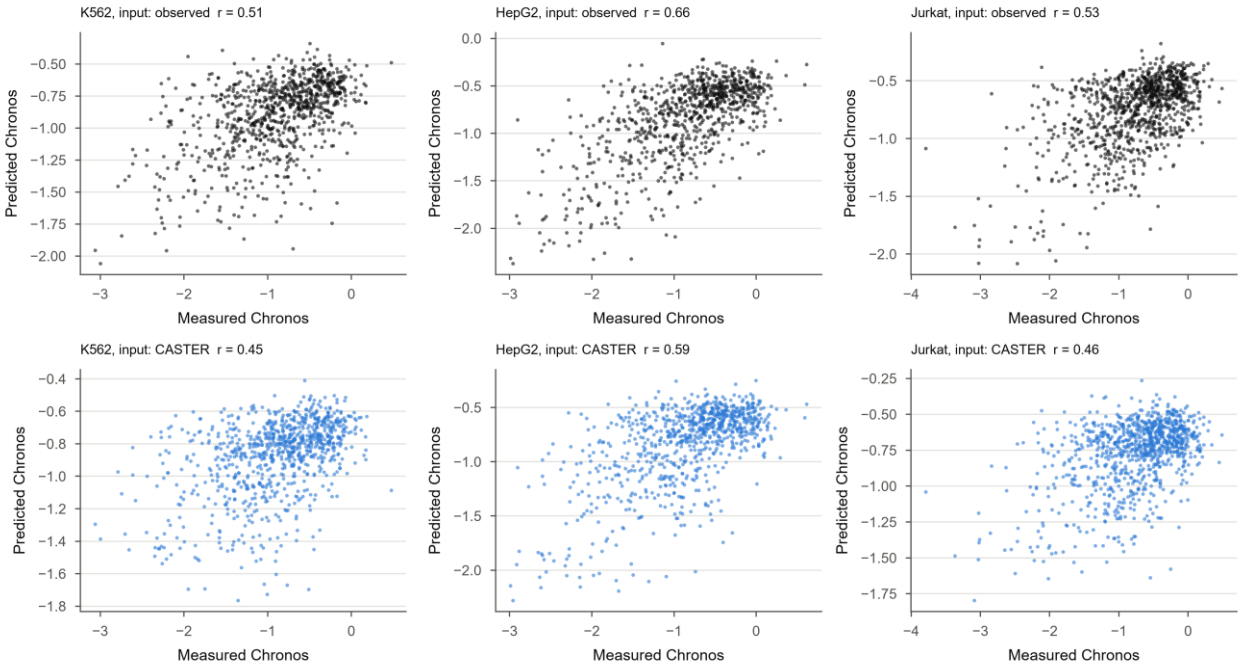


Figure S5. Dependency prediction from transcriptional shifts, related to Figure 5

Measured versus predicted Chronos gene effect per cancer line, from observed shifts (top) and from CASTER-predicted shifts (bottom).

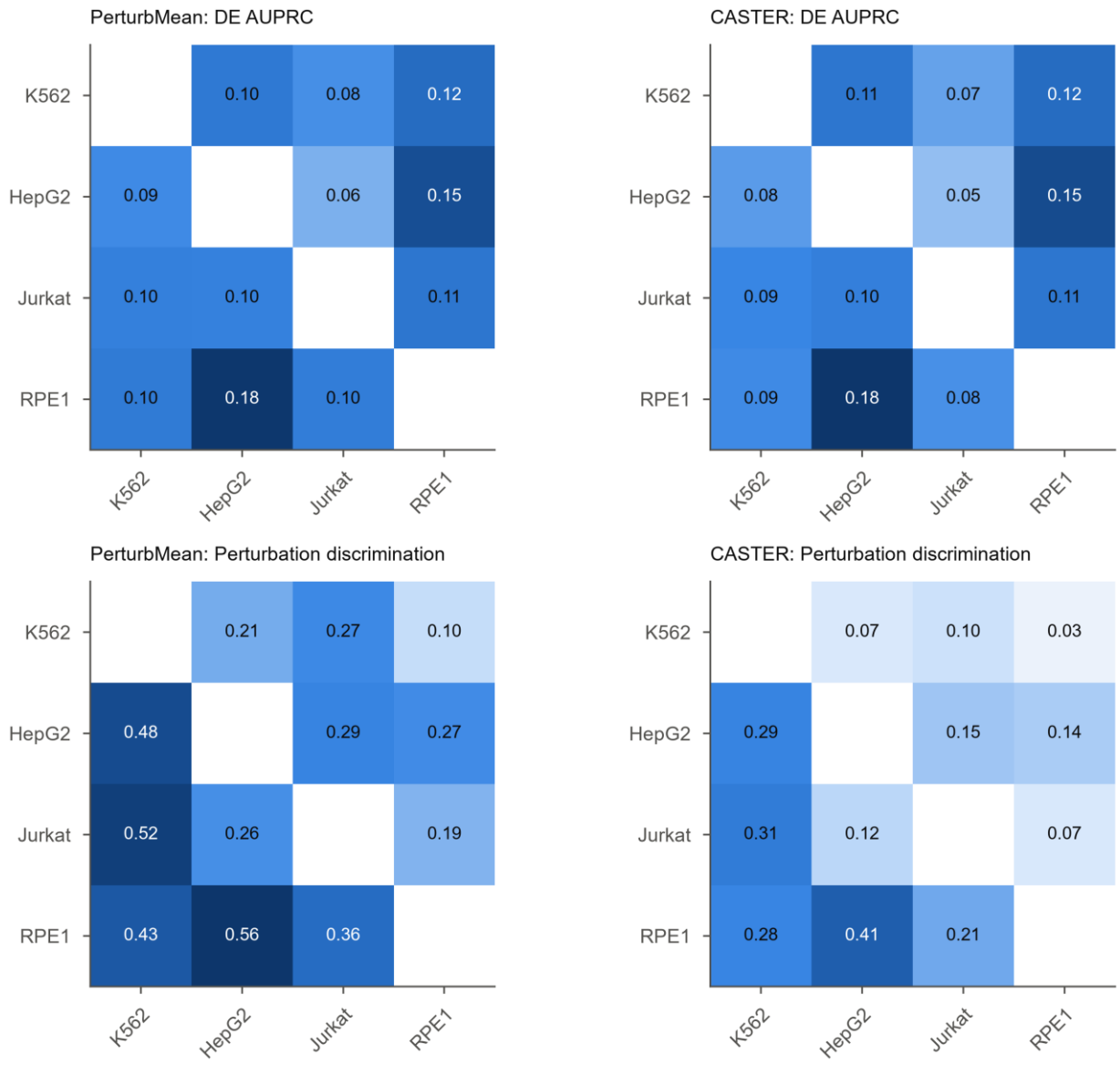


Figure S6. Single-source transfer, all metrics, related to Figure 3

DE AUPRC and perturbation discrimination for every source and target pair.

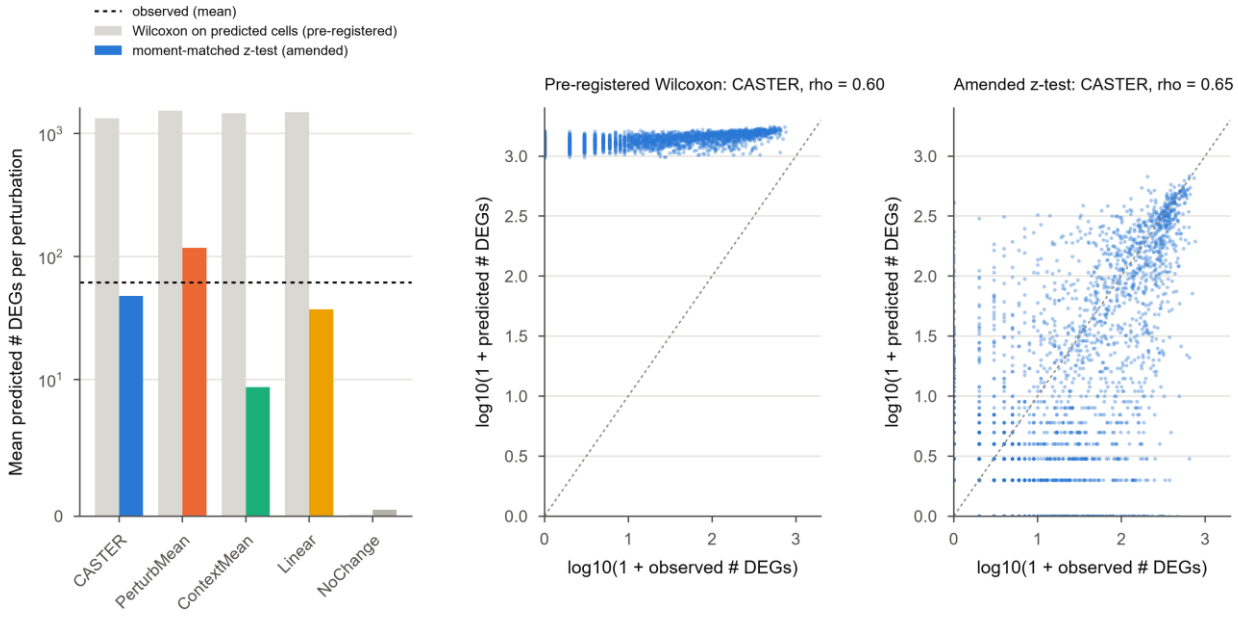


Figure S7. Predicted-side differential expression: pre-registered versus amended test, related to STAR Methods (AMENDMENT-002)

(Left) Mean number of predicted DEGs per perturbation under the pre-registered Wilcoxon test on predicted cells (grey) and the amended moment-matched z-test (colour); dashed line, observed mean. Shifted predictions turn exact zeros into small positive values, so the rank test calls most genes significant; unshifted control cells (NoChange) are unaffected. (Middle, right) Observed versus CASTER-predicted DEG counts per perturbation under the two tests.

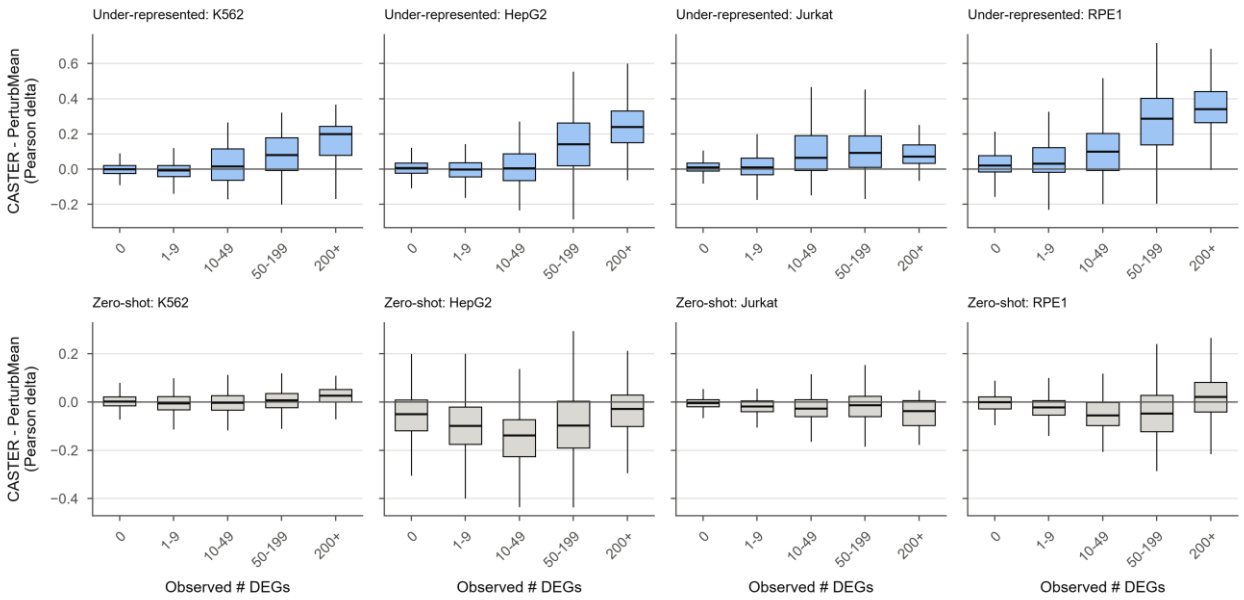


Figure S8. CASTER's gain over PerturbMean as a function of response strength, related to Figures 2 and 3

Per-perturbation difference in Pearson delta (CASTER minus PerturbMean), binned by the observed number of DEGs, for the under-represented (top) and zero-shot (bottom) tasks. Boxes, IQR; whiskers, 1.5x IQR.

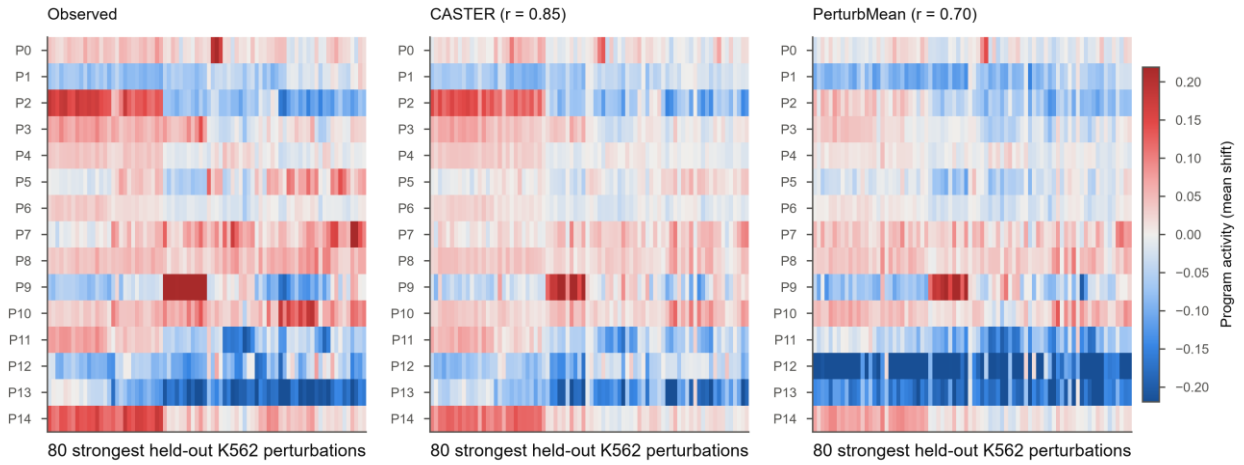


Figure S9. Gene-program activity of held-out perturbations, related to Figure 5

Activity of the 15 K562 gene programs (rows) for the 80 strongest held-out K562 perturbations (columns, clustered on the observed matrix): observed, CASTER and PerturbMean. r, Pearson correlation with the observed matrix.

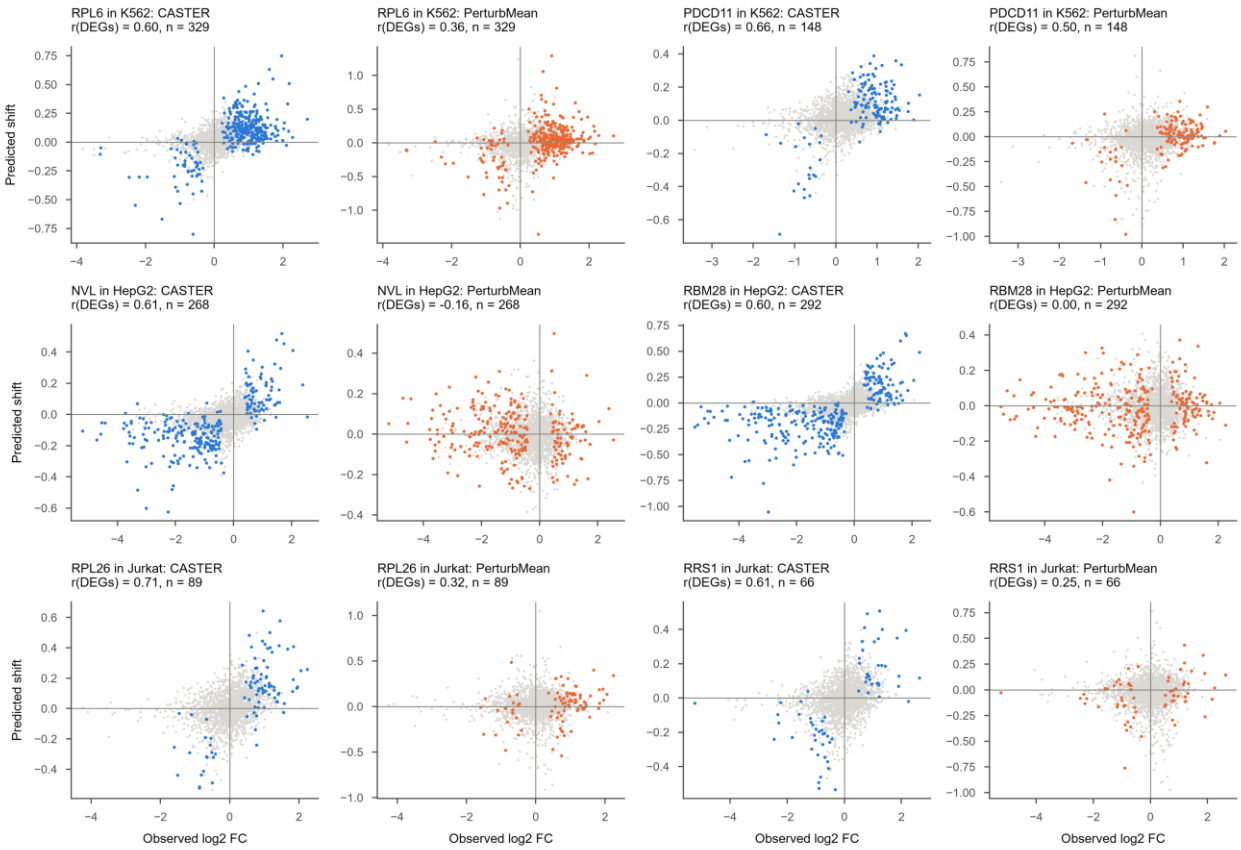


Figure S10. Context-specific example knockdowns, related to Figure 4

For each cancer line, the two knockdowns (>= 30 DEGs) with the largest CASTER gain over PerturbMean: observed log2 fold change versus predicted pseudobulk shift; coloured, observed DEGs.

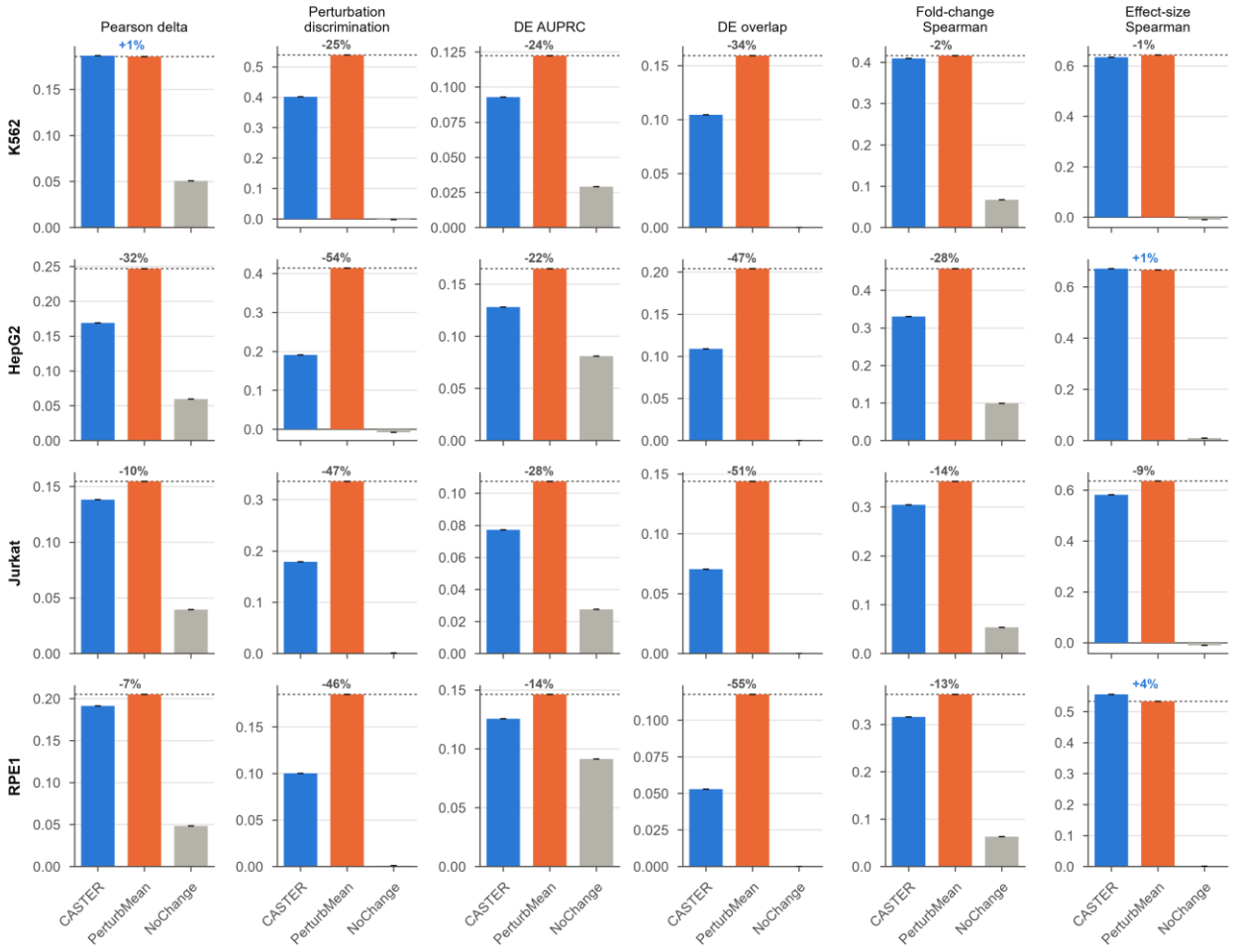


Figure S11. All metrics per held-out context in the zero-shot task, related to Figure 3

As Figure 2C, for the zero-shot task (CASTER, PerturbMean and NoChange).

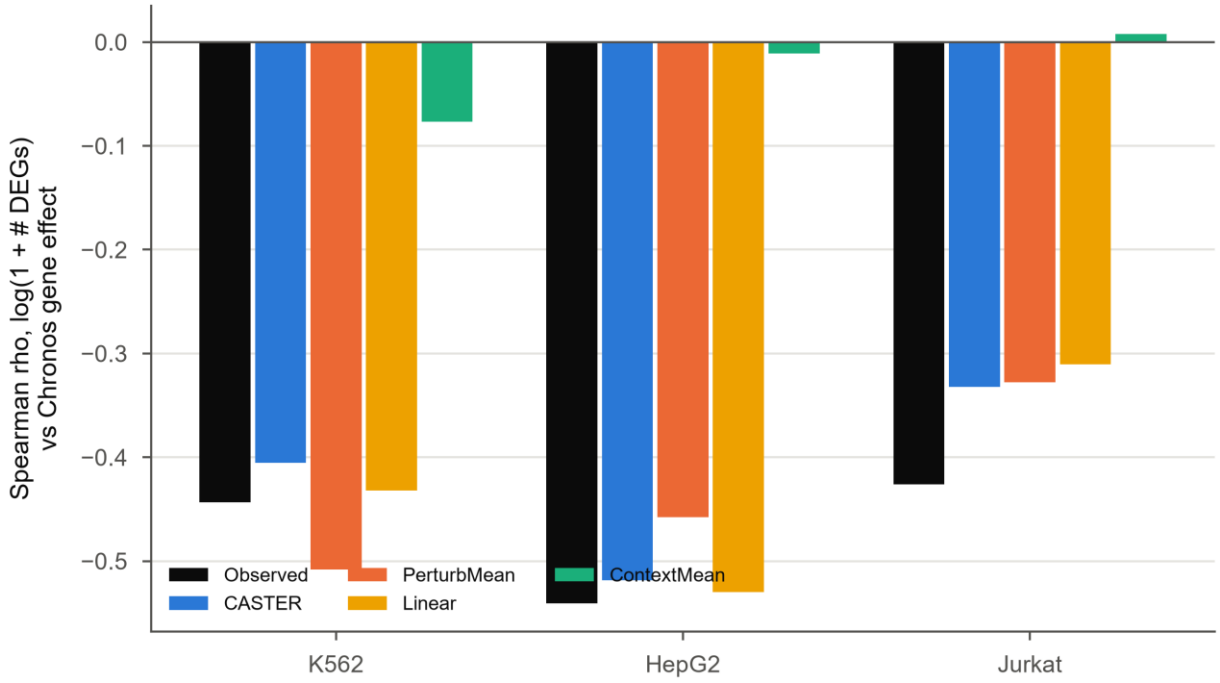


Figure S12. Dependency-response coupling in predicted responses, related to Figure 5

Spearman correlation between response magnitude (log(1 + # DEGs)) and DepMap Chronos gene effect for observed responses and for the responses predicted by each method (held-out perturbations, under-represented task).